

BLOCK 3
UNDERSTANDING:
ENVIRONMENTAL ISSUES

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Introduction

Unit 10: Introduction to Climate Change

The journey begins with an exploration of the fundamentals in "Introduction to Climate Change." We delve into the definition of climate change, unravel its complex causes, assess the extent of its impact, and critically examine the policy implications. This unit serves as the cornerstone for comprehending the broader context of environmental issues, laying the groundwork for the subsequent units.

Unit 11: Climate Change at Global Level

Venturing into the global arena, "Climate Change at Global Level" takes us on a historical voyage through the climate change debate. We scrutinize various protocols related to climate change, with a special focus on the Kyoto Protocol. By understanding the global landscape, we gain insights into the collaborative efforts and challenges faced by nations in addressing this pressing issue.

Unit 12: India's Policy on Climate Change

Shifting our focus to the Indian perspective, "India's Policy on Climate Change" unravels the intricacies of India's stance on this global challenge. We explore key events such as the Copenhagen Summit and the Paris Agreement, providing a comprehensive understanding of India's commitments on the international stage. The unit concludes by delving into the National Action Plan on Climate Change, highlighting the country's proactive measures to mitigate environmental impact.

Unit 13: State Level Action Plan

Zooming in further, "State Level Action Plan" scrutinizes the efforts made by state governments and agencies in combatting climate change. By examining the intricacies of local initiatives, we gain a holistic perspective on the collective efforts to address environmental concerns. This unit serves as a bridge between global policies and grassroots actions, emphasizing the importance of localized responses to a global crisis.

As we conclude this block on "Understanding Environmental Issues," we invite you to reflect on the intricate web of challenges posed by climate change and the role of media studies in disseminating crucial information. Through a journey that spans global debates, national policies, and grassroots initiatives, we hope to empower you with the knowledge and insights needed to engage with and contribute to the ongoing discourse on environmental sustainability. Embrace the opportunity to critically analyze, discuss, and shape narratives that play a vital role in addressing one of the most pressing issues of our time.

UNIT 10 INTRODUCTION TO CLIMATE CHANGE

Structure

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10.0 INTRODUCTION

Climate change is not new for the planet earth. It has been happening across history. We have reasons to believe that during the past 650,000 years there have been seven cycles of glacial advances and retreat. The last ice age ended abruptly about 11,700 years ago and marked the beginning of the modern climate era and human civilization (NASA, 2020). Most of such climatic changes can be attributed to small variations in the earth's orbit that change the incoming solar energy to our planet.

As per the report from the scientific working group in the IPCC TAR (**IPCC, 2001**) the Earth's climate system has demonstrably changed on both global and regional scales since the pre-industrial era. This phase of climate change carries a special significance because it is extremely likely (with 95-100 % probability) due to human activities (IPCC, 2014, p4). There is compelling evidence based on observations, theories and modelling that much of the climate change which we are seeing at the moment is due to human activities. Such anthropogenic activities have been progressing at an unprecedented rate since the onset of industrial societies. This period which is quite often referred to as a period of great acceleration coincides with the accelerated rate of emission of Green House Gases (Steffen et al., 2015). There is enough

scientific evidence to prove a positive correlation between GHGs and temperature (Petit et al, 1999, p429). According to the analysis of 11,944 scientific papers, written by 29,083 authors and published in 1980 journals, there is over a 97% consensus on anthropogenic global warming (Cook et al. 2013; Lehtonen, Salonen and Cantell, 2019). However, there is still major uncertainty about the magnitude of the rise in Global Mean Surface Temperature (GMST) in response to a given amount of greenhouse forcing.

In the past century alone, the temperature has increased by 0.7 degrees Celsius, which is roughly ten times faster than the average rate of ice age recovery warming (NASA, 2020). The technological advances, particularly in the field of palaeoclimatology have now significantly enhanced human capacities to monitor changing climatic conditions on earth, analyze the data and reveal the signals of changing climate.

Current global temperatures are 0.8°C higher than pre-industrial levels and research shows that 1.5°C of warming is already locked into the Earth (World Bank Group, 2014a). The business-as-usual approach will lead the world to warming of 2°C by the middle of the century and 4°C or more by the time today's teenagers reach their 80s (World Bank Group, 2014a). Climate change is not merely confined to GHGs and a rise in temperatures but has widespread implications for human existence. This unit aims to give detailed information about definition, causes, mechanism, impacts, extent, intensity and policy.

10.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- define climate change;
- explain the mechanism and causes of climate change;
- explain the extent and intensity of climate change through some case studies;
- explain the impacts of climate change and;
- explain the implications of climate change on the policies.

10.2 DEFINITION OF CLIMATE CHANGE

As per the UNFCCC, climate change is referred to as “any change in climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods” (UNFCCC 1992). However, according to the Intergovernmental Panel on Climate Change (IPCC), an international entity of United Nations (UN), climate change refers to “a change in the state of the climate that can be identified (e.g., by using statistical tests) by changes in the mean and/or the variability of its properties and that persists for an extended period, typically decades or longer” (IPCC 2015, p. 120). So, climate change is “*any change in climate over time, whether due to natural variability or as a result of human activity*” (IPCC 2014). There is a difference between the usage of the term climate

change in the UNFCCC and IPCC. Though the definition given by UNFCCC talks about only human-induced climate change, we find that the IPCC has adopted a broader definition that includes both natural and human-induced climate change.

10.3 CAUSES OF CLIMATE CHANGE

Now, there is categorical research evidence to prove that climate change is real and that human interventions are the major contributing factors to such changes (IPCC, 2014, p4). Though there have been natural climate variability due to natural internal processes within the climate system, there is an overriding impact of alterations in the atmosphere due to human or anthropocentric activities. In a report published in 2015, IPCC had emphasized the impact of human influence saying that “*Human influence on the climate system is clear, and recent anthropogenic emissions of greenhouse gases are the highest in history. Recent climate changes have had widespread impacts on human and natural systems*” (IPCC 2015, p. 2). Another recently published IPCC report highlights that “*human influence has become a principal agent of change on the planet, shifting the world out of the relatively stable Holocene period into a new geological era, often termed the Anthropocene*” (Allen et al. 2018, p. 53)

Response to climate change has changed both in definition and scope. Though initially, the focus was on reducing and stabilizing human-induced GHG emissions in the atmosphere, it has subsequently become broad-based (UNFCCC 1992). In addition, the scope of climate actors has also started involving non-state actors like the private sector, cities, civil society, and multilateral development institutions among others (UNFCCC 2018).

10.4 DRIVERS OF CLIMATE CHANGE

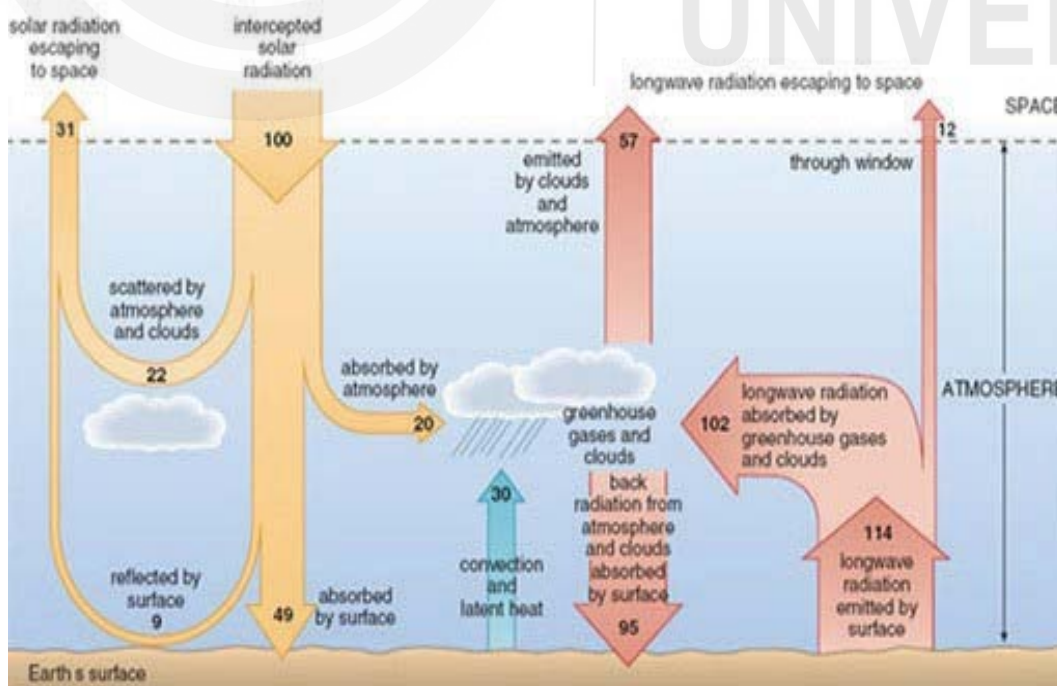


Fig. 10.1 Energy Budget of Earth and its atmosphere: A Schematic representation
Source: UKOU website

The schematic diagram as shown in Fig. 10.1 displays the energy budget of the earth and atmosphere. The figures are global annual averages expressed as a percentage of the rate per unit area at which solar radiation is intercepted by the Earth; i.e., 100 units are equivalent to 342 W m^{-2} . Fig. 10.1 demonstrates that the entire Earth-atmosphere system is in a dynamic equilibrium. However, as we can see from this figure that energy circulates within the system at a higher rate compared to the rate of input or output at the top of the atmosphere. Therefore, we find the earth's surface to be warmer than it otherwise would be. We also observe from this figure that there is no net accumulation of thermal energy in any specific part of the earth-atmosphere system. Therefore, Global Mean Surface Temperature (GMST) cannot change in this scenario. It is however interesting to find out what might cause Earth's GMCT to change? If we look at the energy budget as depicted in Fig.10.1, we can easily understand that any disturbance in the radiation balance at the top of the atmosphere has the potential to force the global climate to change. Such a disturbance can either warm-up or cool down till a balance is reached. This kind of energy perturbation in the energy balance of the entire earth-atmosphere system (expressed in W m^{-2}) is called radiative forcing. Starting from its first major report published in 1990, IPCC has used the concept of "radiative forcing" as the potential mechanism for climate change. There could be several ways in which the energy balance can get disturbed.

1. Change in Solar Constant

For instance, we can work out the effect of change in the solar constant. Let us assume that planetary albedo remains unchanged (at 31%). Under these circumstances, an increase in the solar constant will lead to a positive radiative forcing. It means that the earth-atmosphere system will absorb solar radiation at a higher rate than the rate at which it emits. It will have a warming effect. Similarly, a reduction in solar constant will create a negative radiative forcing thereby leading to a cooling effect.

2. Volcanic Eruptions

The energy balance can also be disturbed by volcano eruptions which generally discharge a huge quantity of gases and volcano ash into the atmosphere. Sometimes these eruptions are so powerful that they inject materials up to the stratosphere. At such high altitudes, these materials gradually spread around the world and bring about a widespread cooling effect for a long time. The sulphur dioxide emitted during the eruption has a far-reaching impact as chemical reactions convert this gas into droplets of sulphuric acid. These sulphate aerosols persist in the stratosphere for several years and enhance the backscattering of solar radiation. However gradually these aerosols comedown to the troposphere and get washed away by rain.

BOX- 10.1: Case Study: Impact of Volcano Eruption on Climate Change

The eruption of Mount Tambora, a large volcano event in Indonesia in April 1815 and its impact on climate change is well documented. It resulted in a

period of abnormally cold weather during the spring and summer of 1816 and its impact was severely felt in many parts of the Northern Hemisphere, especially in the northeastern United States. Average temperatures in New England were observed up to 3.5 °C below normal in June besides unseasonal frosts and snowfalls. Similarly, Europe also suffered from crop failures and famine in England, France and Germany. The below-average temperatures lasted for about two years. Even in the summer of 1816, there were also widespread reports of a dim Sun or persistent haze that was not dispersed by surface wind or rain (since it was actually up in the stratosphere).

3. Accumulation of Greenhouse Gases in the atmosphere

The increasing burden of greenhouse gases can also create imbalances in the radiation budget of the earth-atmosphere system and thus give rise to radiative forcing. This fact can be well understood through an illustrative example. Let us assume that the atmospheric concentration of carbon dioxide is doubled instantaneously but other parameters, for instance, the solar input, planetary albedo, concentrations of other greenhouse gases, etc. remains the same. The obvious impact will be higher absorption of outgoing long waves and consequent reduction in net emission to space (from 236 W m^{-2} to 232 W m^{-2}) for a CO_2 -doubling. It gives a very reliable reason to believe that increasing anthropogenic concentrations of greenhouse gases have created warming of the earth's atmosphere (which may also be referred to as greenhouse forcing).

Observed warming in the earth's atmosphere cannot be explained keeping in view the natural factors alone and thus can largely be attributed to anthropogenic influence. IPCC (2007) states that during the last 50 years "the sum of solar and volcanic forcings would likely have produced cooling, not warming".

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. List out three possible factors which could disturb the radiation balance at the top of the atmosphere?

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10.5 EXTENT OF CLIMATE CHANGE

We have sound reasons to believe that the concentration of carbon dioxide and other greenhouse gases is increasing in the atmosphere. Furthermore, monitoring programs established during the 1980s reveal an upward trend in the levels of two other natural greenhouse gases namely methane (CH_4) and nitrous oxide (N_2O). There are very convincing reasons to believe that

buildup of all three greenhouse gases is non-linear and can be attributed to human intervention including a very unlikely source - the vast ice sheets of Greenland and Antarctica.

It is now well-realized and understood that fossil-fuel-based material consumption and changing lifestyles are the major culprits for climatic change. These factors have increased the emission of Greenhouse Gas (GHG) for example carbon dioxide, methane, and nitrous oxides (Allen et al. 2018, p. 53). The impact of such emissions results in rising temperature across the globe which has manifestations like increased floods and droughts, sea-level rise, biodiversity loss, loss of agricultural productivity, and risks to human health. Such impacts thwart initiatives for sustainable development, particularly in those countries which are most vulnerable to its impacts. The extent of climate change as manifested in different indicators is summarized in Table 10.1.

Table 10.1: The Extent of Climate Change as manifested in different indicators

Indicator	The extent of climate change
Levels of CO ₂ observed at Mauna Loa	The data collected from Mauna Loa shows a consistent rise in CO ₂ concentration. These measurements began in 1958 when the level was 315 ppm. However, levels had reached more than 378 ppm in 2004. It appears to be accelerating. For instance, it took around 200 years (1750 to 1975) for carbon dioxide levels to build up from 280 to 330 ppm but it took just 30 years for it to increase by a further 50 ppm. It has also been seen that the three natural greenhouse gases are fairly uniformly mixed up in the troposphere. Hence their concentration in the atmosphere does not vary much from place to place.
Rising global mean temperature	Global mean warming is now approximately 0.8 degrees Celsius above preindustrial levels (World Bank,2012, ch2, p6)
Increasing ocean heat storage	Approximately 93 per cent of the additional heat absorbed by the earth, resulting from increasing greenhouse gas concentration since 1955 is stored in the ocean. Between 1955 and 2010 the world's oceans to the depth of 2000 m have warmed on average by 0.09 degrees Celsius (World Bank,2012, ch2,p7).
Rising Sea Levels	Sea level rise has been indicated to be 20 cm from the pre-industrial times to 2009. The rate of sea-level rise was close to 1.7mm/year during the 20th century. This accelerated to 3.2 mm/year on average since the beginning of the 1990s (World Bank,2012 ch2, p8).
Increasing loss of ice from Greenland and Antarctica	Both the Greenland and Antarctic ice sheets have been losing mass since at least the early 1990s and losses of ice are shared roughly equally between them (World Bank,2012 ch2, p8).

Loss of Arctic Sea Ice	The linear trend of September ice extent since the beginning of the satellite record indicates a loss of 13 per cent per decade (World Bank,2012 ch2, p12).
Ocean Acidification	From 1750 to 1994, a decrease in surface pH of 0.1 has been calculated, corresponding to a 30 per cent increase in H ⁺ ions. An increase in ocean acidity is more pronounced in higher latitudes than in tropics or subtropics (World Bank,2012 ch2, p11).
Heatwaves and extreme temperature	Heatwaves with temperatures typically more than 3 Standard Deviation warmer, which were highly unusual earlier, started getting more frequent. The five hottest summers in Europe since 1500, all occurred after 2002(World Bank,2012 ch2, p13).
Source: Compiled from World Bank (2012, ch 2)	

BOX 10.2: The CO₂ Measurement Observatory at Mauna Loa

The hypothesis proposed by John Tyndall (Tyndall 1861) that the concentration of atmospheric CO₂ influences the climate of the earth, were systematically investigated by Charles D. Keeling. He started systematically measuring atmospheric CO₂ in 1958 at the Mauna Loa Observatory, Hawaii (Keeling et al. 1976; Pales & Keeling 1965). Mauna Loa was especially identified as a suitable place for this purpose primarily because it was remote from external sources and sinks of carbon dioxide. Mauna Loa Observatory is located on the north flank of Mauna Loa Volcano, on the Big Island of Hawaii, at an elevation of 3397 meters above sea level. The Observatory has been collecting and monitoring data related to atmospheric change since the 1950s. Mauna Loa is strategically identified for measuring CO₂. Firstly, it's isolated and far from sources of pollution and secondly, its lava coated flanks are free from plants and trees thereby making it free from their processes of photosynthesis. Despite its location on the active volcanic island, volcanic emissions, from the summit do not reach the observatory where atmospheric carbon dioxide is measured. However, any rare instance of elevated level can be easily removed from the final data set using simple mathematical filters (Earth Observatory, 2006). The measurements of Keeling show an increase from 316 ppm (parts per million) in March 1958 to 391 ppm in September 2012. However, the seasonal variations were superimposed on this generally increasing trend. Such small variations can be attributed to the growth of plants in the Northern Hemisphere leading to greater uptake of carbon from the atmosphere (Pales and Keeling 1965).

Source: NASA

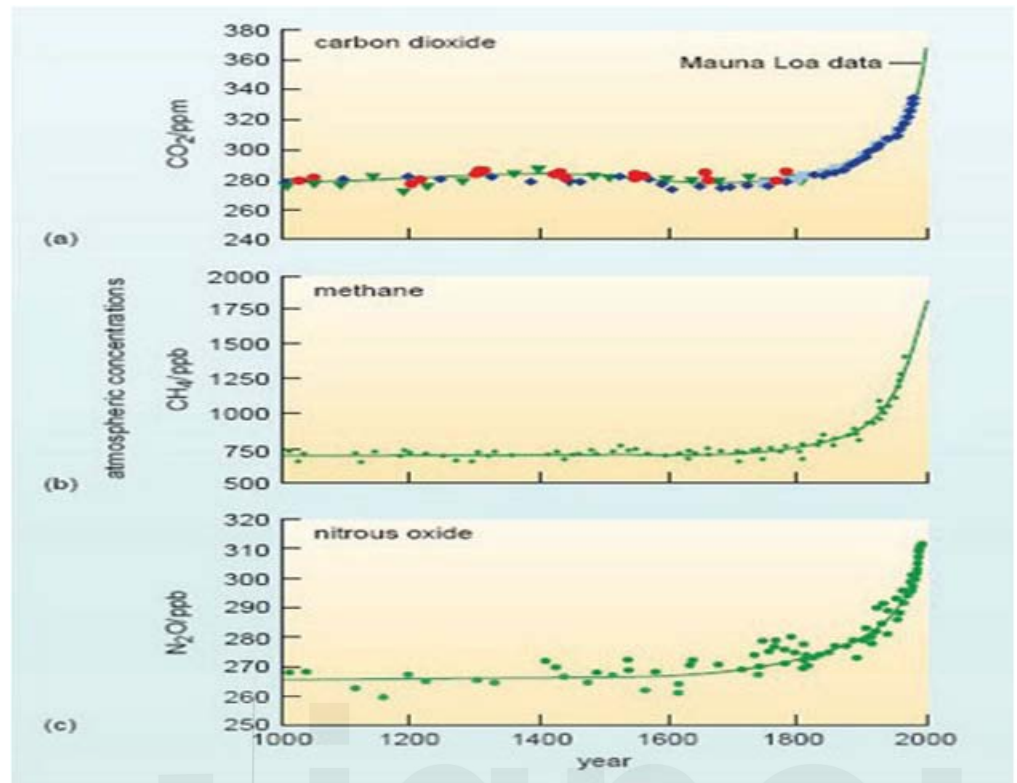


Fig. 10.2: Changing concentration of Greenhouse Gases

BOX 10.3: HOW DO WE MEASURE TEMPERATURES PREVAILING

IN PAST CLIMATES

It is a challenge to find out the temperatures prevailing in past climates. There are several proxy ways to reconstruct past climates. These methods give us indicative figures of the temperatures.

a. Archaeological Inscriptions and Historical Documents

For a long time in human history, people have recorded the vicissitudes of climate and their impact on human affairs. Such archaeological inscriptions and historical documents (diaries, ship logs, etc.) are a valuable source of information about prevailing climatic conditions.

b. Palynology

Plants produce pollen in large quantities which are extremely resistant to decay. Such pollens can be found in sediment layers, the density of which gives indications about prevailing climatic conditions (palynology). The abundance of the pollens of a given vegetation period or year depends partly on the weather conditions of the previous months.

c. Study of ancient ice cores (Paleoclimatology)

Several proxy climate indicators like oxygen isotopes, methane concentrations, dust content as well as many other parameters are used in palaeoclimatology to examine ice core data. For instance, the ratio between the ¹⁶O and ¹⁸O water molecule isotopologues in an ice core gives indicative figures of past temperatures. The heavier isotope condenses more readily as temperature decreases whereas the lighter isotope needs colder conditions to precipitate. Hence analysis of isotope

ratios in ancient ice provides information about climatic conditions in the past. Similarly, air bubbles in the ice, which contain trapped GHGs for example carbon dioxide and methane also give indications about past climatic changes. The European Greenland Core Drilling Project drilled in central Greenland from 1989 to 1992. The ices were found to be 3840 years old at a depth of 770 m, 40000 years at 2521m and 200000 years old or more at 3029 m bedrock. Similarly, the ice cores in Antarctica can reveal climate records for the past 650000 years.

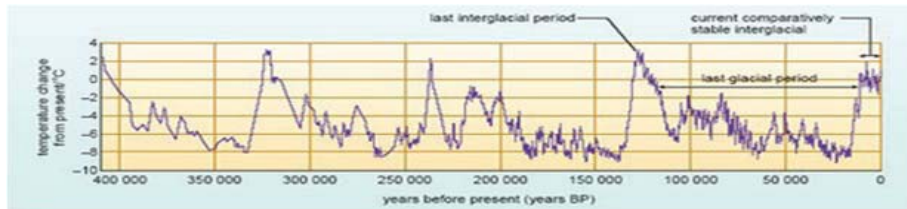


Fig. 10.3: Temperature changes over the past 400 000 years reconstructed from the Vostok ice core, the longest continuous ice-core record to date.

e. Determining past climate from annual tree rings (Dendroclimatology)

Dendroclimatology is the science of determining past climate from annual tree rings. The trees in many parts of the world experience an annual growth cycle. The tree rings are wider when conditions favour growth and narrower during difficult times. The thickness and/or density of a ring depends on the local temperature and moisture conditions thereby giving a unique opportunity to find the temperatures of the respective climate period. There are other properties like maximum latewood density (MXD) which are better proxies than simple ring width. This data can then be matched with overlapping records from other trees to produce longer time series.

Using tree rings scientists have estimated many local climates for hundreds to thousands of years. In a few cases, the preservation of fossil trees has allowed continuous records from 11 000 years ago to the present to be constructed.



Fig. 10.4: The growth pattern from year to year appears as a series of rings that give valuable information about the respective climate period. This science is called *dendroclimatology*

f. Ocean Coral Skeletal Rings or bands

Palaeoclimatological information can also be derived from Ocean Coral Skeletal Rings or bands. Cyclical responses lead to annual banding in corals, which can provide information about sea-surface temperatures, sea level and other ocean conditions - typically back to some 400 years ago.

Source: UKOU

10.6 IMPACT OF CLIMATE CHANGE

Climate change has widespread impacts extending well beyond an increase in temperature. It has an implication for almost everything which we depend upon, for example, water, energy, transportation, wildlife, agriculture, ecosystems, and human health. Anthropogenic activities have already caused major climatic changes and have set in motion, still more changes. Moreover, carbon dioxide, the predominant Greenhouse Gas (GHG), continues to linger in the atmosphere for hundreds of years and it takes a while for the planet to respond (for example oceans). As a result of this, the impact persists for at least several more decades, even if we stop emitting. In the absence of major action to reduce emissions, global temperature is on track to rise by an average of 6 °C (10.8 °F), according to the latest estimates (NASA Global Climate Change). The temperature of the earth's surface does not react instantaneously to the rising carbon dioxide levels. It happens because a sizeable amount of excess energy is stored in the ocean (large heat capacity). Even if all emissions were to stop today, some further warming will be unavoidable before temperatures stopped rising. Therefore, we cannot afford to wait to control GHG emissions till it becomes intolerable.

The warning signs of climate change have started impinging upon the lives of people (Please see Box 10.4). The battle against climate change will have to be fought on two major fronts 1) “mitigation” – reducing the flow of greenhouse gases into the atmosphere; and 2) “adaptation” – learning to live with, and adapt to, the climate change that has already been set in motion. It will require both a globally-coordinated response (such as international policies and agreements between countries, a push for cleaner forms of energy) and local efforts on the city- and regional level (for example, public transport upgrades, energy efficiency improvements, sustainable city planning, etc. (NASA Global Climate Change).

10.6.1 Impacts of Climate Change on Atmosphere

All these anthropogenic changes have the potential to modify the chemical composition of the atmosphere. Changes in atmospheric temperature also affect the rates at which chemical reactions take place (World Meteorological Organization, 2009). Climate warming is expected to enhance the release of biogenic hydrocarbons into the atmosphere which in turn contributes to the worsening of regional air quality (World Meteorological Organization, 2009). Emissions of nitric acid by bacteria in soils are sensitive to temperature and soil moisture and are affected by climate change (World Meteorological Organization, 2009). An increasing number of wildfires and droughts will lead to larger emissions of combustion products like carbon monoxide, nitric oxide, soot and other compounds, affecting the regional and global air quality. Changes in ocean temperature affect the ocean-atmosphere exchanges of compounds such as dimethyl sulphide, which are a source of sulphate aerosols. Just like greenhouse gases, aerosols can have profound impacts on the earth's radiative and energy balance. By controlling the amount of energy that reaches the earth from the sun, aerosols can create a cooling effect by reflecting the sun's energy into space. However, there is one aerosol soot (also known as black carbon), that can create a warming

effect and thus contribute to global warming. Climate models have predicted that aerosols have masked about 50 per cent of the warming that would otherwise have been caused by greenhouse gases (NASA, 2009). Intergovernmental Panel on Climate Change has concluded that industrial aerosols have acted as a significant brake on the increase in global temperatures over the last 30 years or so. Although aerosols have limited the warming, they also have detrimental effects on human health such as lung damage. Aerosols also affect rainfall patterns, some areas like India and China can experience reduced rainfall. They can also alter the pattern of wind and atmospheric circulation. Aerosol as air pollutants cause tens of thousands of premature deaths every year in the UK alone (the University of Leeds, nd).

10.6.2 Heat Waves and Extreme Temperatures

Climate change also leads to more extreme weather events for example more intense hurricanes, more droughts, and more floods, with all the devastation to life, livelihood and property that accompanies them. The past two decades have witnessed a greater occurrence of extreme heatwaves leading to heat-related deaths, forest fires, and harvest losses. Such events include the European heatwave of 2003, the Greek heatwave of 2007, the Australian heatwave of 2009, the Russian heatwave of 2010, the Texas heatwave of 2011, and the U.S. heat wave of 2012. Quite often referred to as 3-sigma events, these events involve monthly and seasonal temperatures typically more than 3 standard deviations (sigma) warmer than the local mean temperature. Normally frequency of such events is only once in several hundreds of years. If we count the five hottest summers in Europe since 1500, we surprisingly find that all of them took place after 2002. The European heatwave of 2003 and the Russian heatwave of 2010 took the exceptional death toll of 70,000 and 55000 respectively. Extreme summer temperatures are gradually affecting an increasing proportion of the land area. They can be attributed to growing climatic warming since the 1960s. In the 1960s, three-sigma events were practically absent and affected less than 1 per cent of the Earth's surface. However, the area affected by such extreme events has grown to 4–5 per cent by 2006–08, and 6–13 per cent of the land surface by 2009–11. The studies have now predicted that such events typically cover about 10 per cent of the land area now. It certainly could not have happened in the absence of global warming.

10.6.3 Socioeconomic Impacts

Climate change can have far-reaching socio-economic impacts. Societies that are most at risk are close to physical and biological thresholds. Ocean warming for example can reduce fishing catch, affecting 650 million to 800 million people relying on fishing for their livelihoods. In Ho Chi Minh City, direct infrastructure damage from a 100-year-old flood could increase from \$200 million to \$300 million today to \$500 million to \$1 billion by 2050. The average share of annual outdoor working hours lost due to extreme heat and humidity could increase from 10 per cent today to 15 to 20 per cent in 2050. Socioeconomic impacts of climate change can have non-linear impacts as system thresholds are breached and knock-on effects.

Poorer regions often have climates that are closer to physical thresholds therefore, they rely more on outdoor work and natural capital and have fewer financial means to adapt quickly. Therefore, these regions are most at risk. However, the impact of climate change has been devastating even for developed countries like the USA which lost some 1.5 per cent of GDP to such weather-related events in 2017 alone. But some regions are likely to gain from climate change for example Canada where crop yields are expected to increase. Also, the rising temperature can boost tourism in northern Europe.

10.6.4 Climate Change Impacts on Health

Climate change affects many of the social and environmental determinants of health such as clean air, safe drinking water, sufficient food and secure shelter. Extreme heat can contribute directly to deaths from cardiovascular and respiratory disease, particularly among elderly people. High temperature raises the levels of ozone and other pollutants in the air that worsen cardiovascular and respiratory disease. Increased levels of pollen and other aeroallergens in high temperatures can trigger asthma affecting around 300 million people. Rising sea levels and extreme weather conditions can destroy homes and other essential medical facilities. Also, variable rainfall patterns can affect the availability of freshwater thus compromising hygiene and increasing the risk of diarrheal disease, which kills over 500 000 children aged under 5 years, every year.

BOX 10.4: Some Emerging Warning Signs of the Climate Change

1. **Glaciers** are contracting at an average rate of 15m per year in the central and eastern Himalayas. If this trend continues, these glaciers are expected to vanish by 2035. This can have serious impingements on the population depending on the glaciers for drinking water supplies etc. Kumbu Glacier, on the popular climbing route to the summit of Mount Everest, has retracted by 5km. The swelling of glacial lakes in Bhutan and the increasing risk of catastrophic flooding is another example of such a phenomenon (ICIMOD, 2019).
2. The thawing of Permafrost in the Arctic region poses a serious threat to the critical infrastructure of the region. This can also pose threat to the utilization of natural resources and sustainable development of the region. Around 70 per cent of the infrastructure located in permafrost is at high risk of being thawed. One-third of Pan-Arctic infrastructure and 45 per cent of hydrocarbon extraction in the Russian Arctic are located in high hazard regions (Hjort et al., 2018).
3. **Sundarbans** are the natural walls for some coastal areas of India and Bangladesh against climate change. But with rising waters and illicit logging, these mangrove trees are nearing their extinction. Therefore, many local people and local species are losing their habitat. Coastal areas are losing their natural defence against extreme climate change phenomena such as floods and cyclones. In 1998 a large flood occurs in Bangladesh, which covered about 70 per cent of the country

(Schwartzstein, 2019).

4. **Advancement** in flowering time for around 385 British plants species by 4.5 days has been observed. This reflects the serious effects of temperature change on the ecosystem, as flowering is a phenomenon that is sensitive to temperature (Fitter & Fitter, 2002).
5. In **Monteverde, Costa Rica**, twenty out of fifty species of Anurans disappeared during a population crash in 1987. These crashes are expected to be the result of demographic changes that affected communities of birds, amphibians etc. These changes are associated with dry mist frequency, which is negatively correlated with sea surface temperature (Pounds, Fogden, & Campbell, 1999).
6. **Antarctic peninsula** Adélie penguin populations have shrunk by 33%, in the past 25 years population has declined by 33 per cent owing to their declining winter sea ice habitat. Adélies depend on sea ice as a resting and feeding platform. They are being replaced by gentoo penguins (a sub-Antarctic species that has begun to migrate towards the pole) which thrive in open water.

Source: Compiled from different Sources

10.7 WHICH COUNTRY HAS CONTRIBUTED THE MOST?

Any country's total carbon emission does not give us a full picture of the country's contribution to global warming. For example, China is the biggest emitter of CO₂ among all the countries (since it overtook the USA in 2007), however, the per capita contribution is just 6.98 tonnes per capita as against 16.24 tonnes per capita of the USA. Sharp variations in CO₂ emission per capita have triggered a debate to fix the responsibility among developed and developing countries. The reasons for such variations are primarily due to the availability of energy resources and the energy policies followed by them. For example, Australia emits around 20.6 tonnes per person which is partly due to its reliance on CO₂ intensive coal. However, consumption recorded in the UK is 11.7 tonnes/person which is partly due to relatively CO₂-light gas power stations. On the other hand, African countries like Kenya have a footprint as low as 0.3 tonnes, which is likely to drop even further with the country's surge in wind power.

BOX 10.5: Global Warming: The Key Contributors

The global average temperature has increased by more than 1°C since its pre-industrial levels. The emission of carbon dioxide has been consistently increasing (~36 billion tonnes of CO₂ per year) and its concentration has reached well over 400 ppm-the highest ever level in over 800,000 years. However, there are marked differences in the contribution of different countries to this massive emission. There are more than 100-fold differences in per capita CO₂ emissions between the countries. Interestingly the countries

in the high- income group emit more whereas those in the low-income group emit less than their population share. As far as the contributions of different countries are concerned, China is the world's largest CO₂ emitter (accounting for more than one-quarter of emissions) which is followed by the USA (15%); EU-28 (10%); India (7%); and Russia (5%). So far USA has cumulatively contributed the most of global emissions (25% of cumulative emissions) followed by the EU-28 (22%); China (13%); Russia (6%) and Japan (4%). There are large inequalities in the emission patterns across the countries which makes climate change a matter of serious international diplomacy. The world's poorest have contributed less than 1% of emissions but are worst affected by the overall impact of emissions. However, a large amount of CO₂ is embedded in traded goods. Therefore, if we look at emissions based on consumption rather than production, figures might change. However, the IPCC guidelines on national emission accounting and reporting are written based on production-based emissions. These are the standards adopted internationally for emission reporting and taken into account for climate reporting. Moreover, such figures are available to the entire world since the mid-eighteenth century. The world is not on track to limit global warming to 2°C above the pre-industrial level, as targeted in Paris Agreement. It is expected that the continuance of existing policies will enhance global warming up to the range 3.1-3.7°C.

Source: Hannah Ritchie and Max Roser (2020): Our World in Data.

10.8 POLICY IMPLICATIONS OF CLIMATE CHANGE

Complex and policy sensitive issues like climate change can evoke conflicting responses from interest groups and policymakers who tend to adopt a “pick n mix” approach to the available scientific evidence, promoting research that reinforces their existing arguments and beliefs, and neglecting or criticizing more uncomfortable findings. Even today, quite often we come across people who deny the climate crisis. It has seriously affected the global fight against climate change. Quite often the facts of climate change come in conflict with people's values, ideologies and political allegiances.

Cooling or warming is not necessarily bad for some areas. For example, Siberia might get warmer in a few decades and this could be profitable for the local population. On the other hand, the impact of climate change may force others to leave their homes because of changing environmental conditions. For instance, in the Sahel region of Africa, for example, declining agricultural productivity has driven people out of their homes for the past 30 years; this is expected to continue as a result of climate change, putting the lives of 60 million people who live in the Sahel region at risk (Goosse, 2005). Further, the opinions of individual scientists are often influenced by decision-makers or the media than the reliability of their knowledge. It has created a scenario where the disputes within the scientific community are often extensively reported by the media and become a matter of public debate.

The politicization of climate change-related issues which have seriously

undermined the scientific pieces of evidence has often been observed. For instance, 2001 shortly after he assumes office, President George W. Bush withdrew the US from the Kyoto Protocol specifying that it would harm the US economy. It was seen across the world as the influence of business-backed lobby groups in government. The allegations were reported in the media that officials of the white house had tried to interfere with a report from US Environmental Protection Agency (EPA) to downplay the seriousness of climate change. Similarly, in 2019 President, Donald Trump who had mocked climate science as a hoax, decided to withdraw from Paris Climate Change Declaration. Though a clearer picture of American participation in the agreement will emerge after the 2020 election, it has created a worldwide concern for the future of global efforts to prevent climate change.

10.9 IMPLICATIONS FOR POST-2015 DEVELOPMENT AGENDA

The Fourth Assessment Report (AR4) of the Intergovernmental Panel on Climate Change (IPCC) unequivocally highlights the rise in global mean temperature and warming of the climate system. Global warming has a widespread cross-cutting impact on almost all the Sustainable Development Goals (SDG). The task of achieving SDGs will be very challenging in a 2°C world, but in a 4°C world there is serious doubt whether this can be achieved at all (World Bank, 2014b). It makes climate change the biggest developmental challenge of our times. The repercussions have already started being felt across the world, particularly in the most vulnerable communities. The situation is serious, as the countries worst affected by climate change are the ones that contribute least to it. As the global temperatures are progressing, problems are likely to further exacerbate, adversely affecting people in every aspect of their lives for instance food availability, water availability, diseases, frequent occurrence of floods, droughts, rising sea levels, oceanic acidification, rising inequalities, etc. It is well realized that strong and early actions based on clean, low carbon pathways are the only answer to slow down the unsustainable growth strategies which will far outweigh the costs. Many of the worst projected climate impacts could still be avoided by holding warming to below 2°C (World Bank, 2014b).

10.10 LET US SUM UP

Climate change presents a complex, borderless and intergenerational phenomenon. It has diverse and far-reaching setbacks to the process of development. As Mike Toman, Research Manager, Research Department, World Bank says “*Climate change is an issue that presents great scientific and economic complexities, some very deep uncertainties, profound ethical issues, and even lack of agreement on what the problem is*”. It is now believed to be a classic example of a super wicked problem. The super wicked problem has been described in the research literature as comprising of four key features: “*time is running out; those who cause the problem also seek to provide a solution; the central authority needed to address it is weak or nonexistent; and, partly as a result, policy responses discount the future*”.

irrationally (Levin et al,2012).

Human civilization is witnessing an unprecedented phase of human history where the availability of natural capital and social capital are the limiting factors for future economic growth (Hawken 1997). Furthermore, the prosperity of one nation cannot be seen in isolation from the prosperity of all other nations. It is well realized that solutions for global problems have to be collective and should be explored simultaneously on social, economic and environmental fronts. Thus, first time in history, human beings can determine the habitability of the planet. The capacities of the people at various levels need to be built to sustain the environment. It is due to this reason that systems of capacity building have come to the forefront of the development agenda. A collective global effort through 'out of box' initiatives is the only answer to address this problem.

10.11 KEY WORDS

Climate System: The climate system is a highly complex system consisting of five major components: the atmosphere, the hydrosphere, the cryosphere, the lithosphere and the biosphere, and the interactions between them. The climate system evolves in time under the influence of its internal dynamics and because of external forcings such as volcanic eruptions, solar variations and anthropogenic forcings such as the changing composition of the atmosphere and land-use change.

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10.13 CHECK YOUR PROGRESS: POSSIBLE ANSWER

Check Your Progress: 1

1. Please refer to section 10.5

UNIT 11 CLIMATE CHANGE AT THE GLOBAL LEVEL

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11.0 INTRODUCTION

In the recent years, climate change as a science has received unprecedented attention from both scientific institutions and the academia. Climate change as an environmental problem is impinging on the every aspects of life including international economy, public health, migration, livelihood, international peace and security. Millions of world's poorest people are already being forced to cope with impacts of climate change. But increased exposure to drought, to more intense storms, to floods and environmental stress is holding back the efforts of the vulnerable population particularly the poor to build a better life for themselves and their children (Atapattu, 2008).

The issue of equity was in the forefront of the climate change debate. The UN Framework Convention on Climate Change and the Kyoto Protocol has reflected the existing disparities between developing and developed countries in creating the environmental problem over a period of time and demanded to develop a legal framework in addressing the challenges of climate change in terms of sharing responsibility at the International level. In this context, climate change is often argued from the perspective of North-South conflict and contestation. Climate change as a global issue is affecting the lives of both developing and developed countries, moreover, the debates on climate change is helping to build up a comprehensive climate change regime with country-specific action plan. The present unit will focus on the historical origin of the climate change debates; Rio declaration on Environment and Development; significance of UNFCCC and IPCC; Kyoto Protocol and Paris Agreement on Climate Change.

11.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- discuss the history of climate change debate;
- discuss the Rio declaration on Environment and Development;
- explain the significance of UNFCCC and IPCC; and
- discuss the Kyoto Protocol and Paris Agreement on Climate Change.

11.2 HISTORY OF CLIMATE CHANGE DEBATE

The development of the climate regime can be traced from the environmental activities of 1980s and 1990s. Climate change as an environmental fact has begun to be felt in the year 1987 with the release of the Brundtland Commission report, *Our Common Future*. The report comprehensively focused on the issues of environmental pollution with regard to depletion of ozone layer, loss of biological diversity and the rising greenhouse gas concentration and argued for achieving sustainability by formulating effective socio-economic policies. Bodansky has divided the development of global climate change regime into five phases prior to the Kyoto protocol. The five periods are as follows;

- The foundational period expresses scientific concern about global warming.
- The agenda-setting phase (1985-88) intended to transform the scientific concern into policy matter.
- Pre-negotiation period (1988-1990) involved government in the process of climate change regime
- The formal intergovernmental negotiation (1992) impressed upon the adoption of FCCC by May, 1992
- The post-agreement phase focused on the elaboration and implementation of FCCC and the adoption of Kyoto commitments.

11.2.1 First Phase: The Emergence of Scientific Consensus

The climate change issue was developed in the scientific sphere with regard to showing concern towards greenhouse gas emission. By the late 1980s, climate change issue was started to discuss in the broader political arena. In the early 1960s, the increasing atmospheric concentration of greenhouse gas was published by the scientists of Mauna, Lao, and Hawaii to establish climate change as a scientific fact. The Keeling curve has accepted as one of the undisputed facts in climate controversy. During 1970s and early 1980s, various computerized models of atmosphere developed and led to increased confidence on the scientists towards prediction of global warming. The 1979 report of the National Academy of Science evaluated the existing models of the atmosphere and concluded that the increase of carbon dioxide in the atmosphere will lead to future climate change. By the mid-1980s, scientists in the World over believe in the fact of anthropogenic emission of other gases like methane and nitrous oxide contribute towards increasing atmospheric concentration of greenhouse gas.

11.2.2 Second Phase: Agenda Setting

The accumulation of scientific knowledge has significantly contributed towards forming both public and political opinion on the issues of climate change. Besides the growth of scientific knowledge, there are other factors responsible for promoting climate change agenda in the international forums. In particular, a small group comprises of scientists working on environmental issues and the chair for IPCC collectively worked to push climate change debate into public. The scientists also worked as knowledge broker while translating scientific facts of anthropogenic emission and further publishing the knowledge about climate variability through channels like conducting workshop and conference, publishing research papers in non-specialists journals and establishing contact with policy makers and political leaders. The establishment of the Advisory Group on Greenhouse Gases and the report of the Enquete Commission helped to translate the uncertainties of climate change science into a real world possibility. The latter half of 1980s was the period which reflected on the rising public concerns on global environmental issues like ozone layer depletion, deforestation, and loss of biological diversity, water pollution and increasing hazardous waste. Incidents like the discovery of Antarctic ozone hole, North American heat waves, and drought of the summer of 1988 were catalytic in raising public consciousness in America and Canada. The Toronto conference in 1988 called for designing a global framework towards protecting atmosphere against crises.

11.2.3 Third Phase: Early International Responses

The year 1988 was marked with the emergence of climate change regime as an intergovernmental matter. The period from 1988 to 1990 was defined as a transitional phase where government has started to play a key role in defending the position of nation-state towards climate change. The establishment of IPCC is a means of reasserting government's position in the climate change issue. The IPCC report which was published in the year 1990

on the scientific assessment of global warming compelled the governments to initiate strong position with regard to climate change concern.

Until 1990s, the government of various Western industrialized countries were primarily involved in the scientific research on climate change and global warming. In the Noordwijk meeting, the split occurred amongst the governments of western countries with regard to committing themselves towards addressing the challenges of climate variability. In this context, many European countries had joined the CANZ group in support of imposing quantitative limitations upon the countries to reduce the emission level of greenhouse gas. In contrast, United States and other Western States continued to give emphasis more on having a neutral position than adopting the European model of target-timetable approach in reducing the rate of greenhouse emission. Developing countries showed unity among themselves and agreed to adopt the target-timetable approach and shown commitment in promoting technology transfer to reduce the emission level. As a Plan of Action, various small island states of developing nations came with alliance and formulate the body, i.e., Alliance of Small Island States (AOSIS) which played a key role in further FCCC negotiations in reducing carbon dioxide emission intensity. On the other hand, the oil producing states critically looked at the science of climate change and argued for adopting a go-slow approach for carbon dioxide emission reduction. Similarly, developing countries like Brazil, India and China emphasized on their right to economic growth and formed a strong opinion to accept the principle of differential responsibility in combating climate change impact. For example, these countries firmly believed that since the problem of global warming was created by the developed countries; therefore the solution to reduce the emission rate may come from the North.

11.2.4 Fourth Phase: Negotiation of the FCCC

In the past decade, the international law has undergone massive transformation with regard to regulating the environment through legal interventions. Prior to 1980s, the air pollution convention in Europe was the only legal machinery addressing the issues related to air pollution and ozone layer depletion. On the other hand, the international customary law did not have specific guidelines concerning the climate change problem. The lack of legal framework to address the climate change problem had demanded for negotiations on a new treaty. According to Bodansky, initially two alternative models of legal framework were developed. The first model was the “Law of Atmosphere” which exclusively focused on addressing the atmospheric pollution in a comprehensive manner. The second model was Vienna Ozone Convention specifically designed for climate change concern. In the year 1990, the UN General Assembly established the body, i.e., Intergovernmental Negotiating Committee for a Framework Convention on Climate Change (INC/FCCC) with specific emphasis on imposing appropriate commitments upon the countries. The primary principles included in the Convention were as follows:

- 1. Target and Timetables:** The European Union and AOSIS advocated to follow target and timetable approach to limit the emission level.

2. **Financial Assistance and Transfer of Technology:** The developing countries demanded for creating a new fund towards helping developing countries in implementing the Convention.
3. **Institution and Implementation Mechanisms:** OECD countries including United States advocated to establish implementation machineries including scientific body, deliberate meetings, reporting, and monitoring of the implementation work to tackle the climate change problem.

11.2.5 Fifth Phase: Post-Rio Developments

Several negotiations were held in the form of meetings with regard to entering into a treaty on climate change. The Convention entered into force in the Berlin conference in the year 1994 to elaborate and implement the commitments to address climate change problem. The Berlin meeting had the following objectives:

- To establish an ad hoc committee to negotiate a protocol;
- To initiate joint-activities among the participating countries to carry out a survey;
- To use Global Environment Facility as the FCCC's financial mechanism; and
- To locate FCCC permanent office at Bonn.

Broadly, the negotiations in relation to climate change were subscribed to two major concerns such as: Emission-target limitations for developed nations; and Establishing different mechanisms for developed nations to meet the targets. The history of climate change regime unfolded both consensus and conflict over issues focused on developing a treaty, partnership of both developing and developed countries, imposing the target-timetable framework on developed nations, and politics of interventions to address the climate change problem. Bodansky summarized the historical pathways of climate change regime as follows:

- During agenda-setting stage, the distinction between government and non-governmental actors got blurred;
- During the stage of actual negotiation of FCCC, the government has taken control to expand its role in combating climate change impacts;
- In the FCCC negotiation, not much clarity was there while explaining each country's position on climate change;
- Kyoto protocol specifies actual obligations for the industrialized countries to limit greenhouse gas emission; and
- In the FCCC negotiations, the debates concerning the target and timetable or financial assistance were taken place more in a political manner over the substantive way.

11.3 RIO DECLARATION ON ENVIRONMENT AND DEVELOPMENT

UN conference on Environment and development met at Rio de Janeiro from 3-14 June 1992. It reaffirmed the declaration of the UN Conference on the Human Environment, adopted at Stockholm on 16th June 1972. UNCED firmly affirmed that development is not an isolated affair. It is part of the environment, economic and social issues. In that conference, the issues of global climate change, biological diversity, deforestation and desertification were debated and many treaties and agreements were signed. And it also stressed that the nation's future policies on economic development should be drafted by considering the environmental implications of socio- economic development. Agenda 21 was a special product of this conference which is also known as Earth summit. Although agenda 21 was not a global consensus document, it is unique in the sense that it created broader perspective on sustainable development and it brought local, national and global debate and action. The preamble states "*Agenda 21 is a dynamic program. It will be carried out over time by the various actors according to the different situations, capacities and priorities of countries and regions. The process marks the beginning of a new global partnership...*". The Earth Summit led to the formation of United Nations Framework Convention on Climate Change (UNFCCC). Now let us learn about UNFCCC.

11.4 UNFCCC

The UN Framework Convention on Climate Change was adopted at Earth summit at Rio de Janeiro in 1992. It is a treaty signed by 192 countries. It does not set any binding targets on countries for cutting emissions. It only calls for stabilizing carbon emissions. But it set in motion international negotiations for settling the issue of who will cut and how much. The parties to the convention have met annually from 1995 onwards in conference of parties (COP). The Conference of the Parties is the governing body of the Convention, and advances implementation of the Convention through the decisions it takes at its periodic meetings.

Table 11.1: COP Meetings

Session	Location	Conference
COP 1	Berlin, Germany	Berlin Climate Change Conference-1995
COP 2	Geneva, Switzerland	Geneva Climate Change Conference-1996
COP 3	Kyoto, Japan	Kyoto Climate Change Conference - December 1997
COP 4	Buenos Aires, Argentina	Buenos Aires Climate Change Conference - November 1998
COP 5	Bonn, Germany	Bonn Climate Change Conference - October 1999
COP 6	The Hague,	The Hague Climate Change Conference -

	Netherlands	November 2000
COP 6 2	Bonn, Germany	Bonn Climate Change Conference - July 2001
COP 7	Marrakech, Morocco	Marrakech Climate Change Conference - October 2001
COP 8	New Delhi, India	New Delhi Climate Change Conference - October 2002
COP 9	Milan, Italy	Milan Climate Change Conference - December 2003
COP 10	Buenos Aires, Argentina	Buenos Aires Climate Change Conference - December 2004
COP 11	Montreal, Canada	Montreal Climate Change Conference - December 2005
COP 12	Nairobi, Kenya	Nairobi Climate Change Conference - November 2006
COP 13	Bali, Indonesia	Bali Climate Change Conference - December 2007
COP 14	Poznan, Poland	Poznan Climate Change Conference - December 2008
COP 15	Copenhagen, Denmark	Copenhagen Climate Change Conference - December 2009
COP 16	Cancun, Mexico	Cancún Climate Change Conference - November 2010
COP 17	Durban, South Africa	Durban Climate Change Conference - November 2011
COP 18	Doha, Qatar	Doha Climate Change Conference - November 2012
COP 19	Warsaw, Poland	Warsaw Climate Change Conference - November 2013
COP 20	Lima, Peru	Lima Climate Change Conference - December 2014
COP 21	Paris, France	Paris Climate Change Conference - November 2015
COP 22	Marrakech, Morocco	Marrakech Climate Change Conference - November 2016
COP 23	Bonn, Germany	UN Climate Change Conference - November 2017
COP 24	Katowice, Poland	Katowice Climate Change Conference - December 2018
Source: https://unfccc.int/process/bodies/supreme-bodies/conference-of-the-parties-cop		

11.5 IPCC

The IPCC was set up jointly by the United Nations' and the World Meteorological organizations and convened its first meeting in the year 1998 to discuss the scientific uncertainties over climate change. The IPCC has described the climate uncertainties in relation to magnitude, timing, and regional political interest among countries over the world to participate in the development of models to address the challenges of global climate change. The IPCC has produced comprehensive assessment reports in 1990, 1995, 2001, 2007, 2014, and 1.5°C special report in 2018. In the preparation of these reports, much debated and contested fact was the climate change of the twenty-first century. In particular, many of the scientists felt that the climate uncertainties are such that it is refraining scientists for making any prediction and estimate for future climate variability (Ibid.). IPCC being an inter-governmental body assured the involvement of government representatives, scientists and policy makers in the process.

The acceptance and realization of the problem of global warming at the global platform has led to the formulation of international Framework Convention on Climate Change (FCCC), which was signed by over 160 countries. After the Earth Summit in 1992, the arguments concerning the scientific importance of IPCC was intensified in the world's press. The Climate Convention signed at the Rio conference acknowledged the importance of the balanced relationship between humans with their environment. The Convention emphasized to maintain the harmonious relationship between human society and their environment towards stabilizing the greenhouse gas concentration in the environment. Further, it set the objective for concrete action to reduce and stabilize the climate change. The signatories to the Convention consensually agreed upon the initiation of mitigation actions to combat the impacts of climate change.

Box 11.1: The Climate Convention

The convention concerned that human activities have been substantially increasing the concentration of greenhouse gases at atmosphere.

- Convention noted that the largest share of global greenhouse gases emission is originated in developed countries as compared to developing countries.
- The convention recognizes that various mitigation actions can be justified economically and can further help in solving other environmental problems.
- The convention specified areas/countries such as low-lying countries, countries with low-lying coastal areas, and semiarid areas or areas liable to flood are vulnerable to climate variability.
- Convention affirmed that action responding to climate change needs to be coordinated with the socio-economic development in an integrated pattern.
- The Convention determined to protect the climate system for both present and future generations.

The primary objectives of the Climate Convention those were contained in the Article 2 are as follows:

- To prevent any anthropogenic interference with the climate system, there is a need to stabilize the concentration of greenhouse gases in the atmosphere.
- The stabilization of the concentration of greenhouse gases needs to be achieved within a specific timeframe while ensuring effective food production and economic development in a sustainable way.

Check Your Progress: 1

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. What are the objectives of Berlin Conference?

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2. What are the principles included in the Intergovernmental Negotiating Committee for a Framework Convention on Climate Change?

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11.6 CLIMATE CHANGE AND THE NORTH-SOUTH DEBATE

The climate change debate needs to address the issue of equity not only in relation to the future generation but also in relation to the present generation. In this context, the main concern of the International Community was the development of an international legal regime which can reflect and address the vast disparities between developing and developed countries with regard to wealth and the greenhouse gas emission. In response to the challenge, the international community phrased the common but differentiated responsibility (CBDR) principle as the guiding principle of climate change science. The principle of differentiated responsibility was well incorporated in the UN Framework Convention on Climate Change.

Common but Differentiated Responsibility

The CBDR principle was derived from the principle of equity, which is an important principle of international law. For the first time, the Rio Declaration on Environment and Development was being instrumental in incorporating the principle of CBDR at the global level. The principle 7 of Climate Convention in Rio Declaration states:

“States shall co-operate in a spirit of global partnership to conserve, protect and restore the health and integrity of the Earth’s ecosystem. In view of the

different contributions to global environmental degradation, States have common but differentiated responsibilities. The developed countries acknowledge the responsibility that they bear in the international pursuit of sustainable development in view of the pressures their societies place on the global environment and of the technologies and financial resources they command” (Atapattu, 2008).

This principle has become the most contested issues in the forum of Rio Conference. Principle 7 often emphasized the legal responsibilities of the developed countries to combat climate change impacts due their contributions towards environmental problems in the past. Often, the principle 7 was debated over the issue of whether the principle is implying legalities or it can be adopted as guiding principle in the Climate Convention. CBDR as the guiding principle was profusely endorsed by the UN Framework for Climate Convention in the Article 3 under ‘principles’. Article 3 states;

“The Parties should protect the climate system for the benefit of present and future generations of humankind, on the basis of equity and in accordance with their common but differentiated responsibilities and respective capabilities. Accordingly, developed country Parties should take the lead in combating climate change and the adverse effects thereof”.

Further, the Article 3 acknowledged the specific needs and vulnerabilities of the developing countries with regard to the impact of climate change and emphasized that developing countries interest needs specific consideration under the Convention as the developing countries parties are bearing the disproportionate burden of climate variability.

11.7 KYOTO PROTOCOL

In the UNFCCC process a treaty was adopted on December 11, 1997 at the Kyoto (Japan COP). The Kyoto protocol as it is known, came into force in February 2005. The most notable exception is the US. Under the protocol, 39 industrialized countries, Annexure I countries, committed themselves to reducing the emissions by 5.2 per Cent of the 1990 levels by 2012. The protocol set up mechanisms for carbon trading. Under this, a country can finance carbon dioxide saving measures in another country and thus offset its own emissions. Similarly a country can buy carbon credits from another that has accumulated them by reducing emissions or through other measures (like forestation). The protocol also set up a fund for helping developing countries adopt to changing climate.

The UN climate change secretariat based in Bonn, Germany is an international transaction log to verify that the transactions are consistent with the rules of protocol. Reporting is done by parties by way of submitting annual emissions inventory and national reports under protocol at regular intervals. A complaint system ensures that parties are meeting their commitments and it helps them to meet their commitments if they need help or if they are in problem. The Kyoto protocol also assists countries in adapting to the adverse effects of climate change. It facilitates the

development and deployment of techniques that can help increase resilience to the impacts of climate change. The adaptation fund was established to finance adaptation projects and programmes in developing countries that are parties to the Kyoto protocol. The fund is financed mainly with the share of proceeds from Clean Development Mechanism project activities. Kyoto protocol applies to industrial nations only. Developing countries like India and China were not required to commit to reductions because their per capita GHG emissions are much lower than those of developed nations. Poorer economies of developing countries are unable to absorb the cost of switching from the fossil fuel based system to cleaner fuel system. Once the cleaner technologies are developed, the poorer countries will actively involve in reduction in carbon emission and climate change agreement.

The Kyoto Protocol was formulated on the basic principle of CBDR and specified that industrialized countries will have to take lead in the process of reduction of greenhouse gas emission. The basic structure of the Protocol and Commitments were agreed at a meeting of the Conference of the Parties in Kyoto in the year 1997 (Ibid). The Protocol embodies substantive obligations in relation to climate change for both developing and developed countries. These commitments ranged from developing national inventories/ strategies of greenhouse gas emissions, formulating and implementing national level programs with regard to mitigation and adaptation strategies of climate change, promoting and cooperating the transfer of sustainable technology, promoting sustainable and integrated development, cooperating in preparing adaptation plans, promoting and supporting scientific research and promoting education, training and public awareness about the effect of climate change (Atapattu 2008). For instance, in the first commitment period from 2008-12, the industrialized countries are committed for specific quantitative reductions in emissions. The Protocol involved inbuilt mechanisms to facilitate concrete action plan for combating climate change and the action plan aimed towards including the developing countries under the general obligations of the Climate Convention.

The Kyoto Protocol involves specific principles and mechanisms to reduce the greenhouse gas emission; however, it does contain any obligations for developing countries with regard to reduction of emission. Therefore, the protocol had led to intense controversy on the issue of equity over climate change. In this context, most of the debates had happened with regard to the principle of general obligations of all the states to cooperate and protect the environment. As Atapattu writes, it is important to recall the wordings of Rio Declaration, i.e., the common obligation of the states to protect the environment. This emphasizes on the two dimensions of the climate change debate such as: global partnership in preserving the environment; and the obligation is common to both developing and developed states with regard to protecting the environment. Kyoto Protocol becomes problematic as it does not contain any differentiated responsibilities for developing countries – on the contrary, it contains no obligations for developing countries for the commitment period (ibid: 8). On the contrary, it was argued that since the Kyoto protocol applies for the commitment period; therefore, no specific obligations were formulated for the developing countries on the basis of their

minimal contributions to greenhouse emission. The exclusion of developing countries from the global climate change regime has reflected the notion of differential principles for developing vis-à-vis developed countries. Further, it was argued that there may be a need to impose differential policies within the developing countries with regard to climate change policies. In this reference, there were alternative apprehensions emerged— many climate change protagonists assumed that in the second commitment period, developing countries may be imposed with the emission reduction obligations similar to the developed countries. Though, both the climate change regime and ozone regime applied the CBDR principles, however, the two regimes treated the principle of differential responsibility in different ways. Similarly, developing nations have shown their commitments towards the global climate change regime differently.

The Kyoto protocol is generally seen as an important first step towards the truly global emission reduction regime that will stabilize greenhouse gases (GHG) emissions and provides essential architecture for any future international agreement on climate change. By the end of the first commitment period of the Kyoto protocol in 2012 a new international framework have to be negotiated and ratified that can deliver the stringent emission reductions as indicated by Intergovernmental Panel on Climate Change (IPCC). Based on this commitment the protocol has evolved five key principles as follows:

1. Commitments
2. Implementation
3. Minimizing impacts
4. Accounting, reporting
5. Compliance

11.7.1 Reduction Target and Ratification Status

Eighty four countries including the United States (US), European Union (EU), Japan, Russia and Canada signed the protocol. According to Kyoto protocol at least 55 countries together producing at least 55% of the world's 1990 carbon dioxide emission must ratify the protocol. Only then the protocol will come into effect. This is known as 55/55 target. In the year 2001, the protocol suffered major setback because US refused to ratify which is producing 36.1 % of the carbon dioxide emission. EU and Japan supported the protocol and ratified it. Reluctance supporters are US, Australia, Russia and Canada. Unlike EU and Japan which are small these countries supply cheap energies also. They felt if protocol is implemented their businesses and economy are in disadvantaged position. But EU, small countries and Japan consider protocol will bring advantages to their economy. The protocol has not been till put into effect due to the non-achievement of 55/55 target. UNFCCC has held annual conference but this did not bring much progress. No penalties exist for non-compliance so far but penalties are under consideration like trade sanctions, financial penalties, emission penalties and any country can withdraw from the treaty of the ratifying it by giving one year notice. This is also criticized. Countries like Canada and Russia which

has large forest cover argued that they should give credits for absorbing greenhouse gases across the globe. Other countries argued that planting of trees as a part of regular industrial projects should also be given credits.

Box 11.2: The Kyoto Mechanisms

The Kyoto mechanisms include three strategies to reduce emission level such as:

- **Joint Implementation (JI)** will allow industrialized countries to facilitate the implementation of projects pertaining to reduction of emission level.
- **The Clean Development Mechanisms (CDM)** allows industrialized countries to implement projects which will reduce the emission level of developing countries. For example, the project on rural electrification by providing solar panels or encouraging reforestation on the degraded land.
- **Emission Trading** allows industrialized countries to purchase assigned amount of units of emission from other industrialized countries to meet the emission target.

(Source: Houghton 2009).

11.7.2 Shortcomings of Kyoto Protocol

The major shortcomings of the Kyoto protocol are:

- Excluding developing countries from the protocol. This will reduce the effectiveness of the agreement.
- Exclusion of India and China which constitute 36% of world's population and both countries are growing economy.
- Using of older, outdated and environmentally unfriendly technologies by the developing countries, which are excluded from the Kyoto protocol.
- Growing unchecked developing countries emission.
- The Kyoto protocol only raises the awareness and not in real terms.

Check Your Progress: 2

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. List out the key principles of Kyoto Protocol.

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2. What are the major shortcomings of Kyoto protocol?

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11.8 MARRAKESH ACCORD

Marrakesh accord deals with capacity building for the representatives of developing countries to ensure their effective participation in the Kyoto protocol. Developing countries have the responsibility to identify their needs and they have to cooperate with other developing countries (south-south cooperation). It also talks about the development and transfer of technologies. COP-7 decided to allot a meaningful and effective framework for technology transfer. The framework was reviewed at the end of 2006 – COP12. It stresses the cooperation between stakeholders including on technology need assessment, technology information, enabling environment, capacity building and mechanism for technology transfer. Global Environment Fund (GEF) and special climate change fund should be made available for financing activities and to organize a workshop on insurance against adverse effects of climate change and implementation of response measures. The workshops on synergies with other multilateral environmental agreements and economic diversification should be organized before COP-9 i.e. before the end of 2003. GEF should stream line procedures and shorten the time between the identification of project and the availability of funding. GEF should fund to Least Developed Countries (LDC) and small island states. The agreement also stressed the need for special climate change fund and LDC fund.

11.9 BALI ACTION PLAN

UNFCCC was held in Bali in December 2007. The Government representatives from both developed and developing countries around the world agreed to step up their efforts to combat climate change and adopted Bali road map. The Bali road map includes Bali action plan which adopted new negotiating process under the UNFCCC with the aim of completing this by 2009. It also includes negotiations under the Kyoto protocol and its 2009 deadline. The Bali action plan adopted by the COP includes long-term cooperative action which goes beyond 2012. The Bali action plan centered on four main building blocks i.e. mitigation, adaptation, technology and financing. It also insisted that future discussion should address the following issues.

1. Measurable, reportable and verifiable nationally appropriate mitigation, commitments or actions by all developed countries.
2. Nationally appropriate mitigation actions by developing country parties supported and enabled by technology, financing and capacity building, in a measurable, reportable and verifiable manner.
3. Other subject for the future discussion include the use of sectoral approaches, approaches to enhance cost effectiveness of mitigation actions including market mechanism and the issue of reducing emissions from deforestation and forest degradation developing countries.
4. During the Bali action plan negotiations, country representatives had negotiations under the Kyoto protocol also.

11.10 COPENHAGEN SUMMIT

Copenhagen accord held in 2009 accepted the need to hold mean temperature increases below 2 degree Celsius. Non annexure 1 countries also pledge to reduce emissions. It also establishes the principles for a system of international monitoring, reporting and verification. The accord paves the way for an increase in future funding for developing countries. The Copenhagen summit had hoped for the effective implementation of international climate change agreements. The European countries and the developing countries are committed to negotiate global climate deal. They strongly believed in universal and comprehensive treaty with the firm commitment for emission reduction stance a change for averting the threat post by global warming. China joined with G77 block of developing countries and demanding a legal binding framework for mitigation by industrialized nation.

Check Your Progress: 3

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. What is the importance of Bali Action Plan?

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2. Write a short note on Copenhagen Summit.

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11.11 PARIS AGREEMENT ON CLIMATE CHANGE

At the Durban Conference (COP 17) an “Ad-Hoc Working Group on the Durban Platform for Enhanced Action” was established “to develop a protocol, another legal instrument or an agreed outcome with legal force under the Convention applicable to all Parties”, by COP 21 so that it can be adopted no later than 2015. Indeed the Paris Agreement is the outcome of four-year negotiating process and multilateral diplomacy. Article 2 of the Paris Agreement, states that “in enhancing the implementation of the Convention, including its objective, the agreement aims to strengthen the global response to the threat of climate change, in the context of sustainable development and efforts to eradicate poverty, including by: (a) Holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to

1.5°C above pre-industrial levels, recognizing that this would significantly reduce the risks and impacts of climate change; (b) Increasing the ability to

adapt to the adverse impacts of climate change and foster climate resilience and low greenhouse gas emissions development, in a manner that does not threaten food production; and (c) Making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development”. Paris Agreement entered into force on 4th November, 2016. The main features of the agreement are universal application; the principle of equity and principle of common but differentiated responsibilities and respective capabilities; and “legally binding agreement that will apply to those states that have expressed their consent to be bound by means of ratification, acceptance, approval, or accession”. With respect to our country, the agreement demands India to submit “national contributions” every five years; and to embark on low carbon development pathway.

11.12 INDIA’S RESPONSE FRAMEWORK

Within the developing nations, India’s around 700 million rural population is dependent on the climate sensitive sectors such as agriculture, forest, fisheries, water, biodiversity and coastal zones. Further, the rural population is more vulnerable to the climate variability as they have low adaptive capacity in relation to extreme and fluctuating weather condition. The National Commission Report found out that the impact of climate change can be felt in sectors of natural ecosystem and the socio-economic system. The IPCC assessment report reveals the following:

- Increase in rainfall over the Indian sub-continent by 6-8 per cent.
- Extreme rise of maximum and minimum temperature is expected in the West Coast of India and West Central India.
- Heat spell have become common in the Northern India.
- There were around 15 major droughts in the past 50 years, which affected the productivity of rain fed crops in drought years.
- Food security is under threat due to occurrence of drought and flood, and variability in the climate.

The adverse impact of climate change has already threatened the existing livelihood in India, for which India has undertaken various sectoral mitigation measures to contribute towards the objectives of United Nations Framework Convention on Climate Change (UNFCCC). India’s development plan is based on the primary principle of sustainability along with inclusive economic growth. There have been initiatives taken in sectors like coal and oil, renewable energy, hydropower, social energy, energy efficiency and conservation, transport, agriculture, power sector, and so on. The current government expenditure on adaptation has exceeded two percent of GDP. The broad areas where the adaptation measures have been carried out include:

- Crop Improvement
- Drought Proofing
- Health
- Risk financing

- Disaster management
- Livelihood Preservation

11.12.1 National Action Plan on Climate Change

India's first Action Plan on Climate Change was prepared in the year 2008 by focusing on the adaptation and mitigation policies and programmes. The plan identifies eight core national missions to be implemented as part of adaptive strategies of climate change in areas of sustainable agriculture, water, Himalayan Ecosystem, enhanced energy efficiency, and accumulation of strategic knowledge on climate change. The plan identifies measures for economic growth, increasing living standard while yielding co-benefits for addressing climate change effectively. Under the economic liberalization policy, the private sectors both domestic and foreign investors are playing key role in developing alternative efficient technologies to deal with climate change. For example, due to import liberalization, joint ventures are possible in providing wide range of consumer goods like energy efficient cars and refrigerators in the market. Consumers are showing preference in the buyers' market and exercising their choice for energy efficiency products. The energy sectors are switching in the direction of using alternative fuel. For example, the power sector is permitted to use natural gas and the coal-based energy plants have become dysfunctional.

In the present scenario, India is actively and decisively participating in the climate change negotiation debates. India's position is not difficult to comprehend, however certain questions have been emerged in the discourse of climate change negotiation. Questions include what concentration level accepted to the developing countries, how do the commitments towards emission reduction will be available to developing countries, how do you ensure that India will have fair share in the global environmental place and so on.

11.12.2 India's Position in the Climate Change Dialogue

In the context of current debate, developing countries like India is taking necessary actions in terms of execution of programmes and action research. Technology transfer needs to take priority in the mitigation and adaptation process. The central argument on the position of India and the other developing countries in the global climate change debate is as follows:

- Developing countries like India in particular have a strong opinion that the unsustainable consumption pattern of the industrialized countries are responsible for the climate change.
- India and other developing countries' economy are vulnerable to climate change and the impact of climate change can hamper the progress in poverty eradication, and the socio-ecological condition of the country.
- Developing countries have potential in limiting GHG emission by facilitating effective programmes like policies aiming towards effective energy and economic management, development of both conventional and renewable energy resources, and limitation of health hazards.

- Developing countries have undertaken certain price reforms and removing subsidies which have positive impact in the context of energy saving.
- With regard to differentiated responsibility, countries like India needs to stabilize the carbon emission level and further aim at reducing the GHG concentration at a sustainable level.
- Developing countries including India needs to adopt an equitable approach while framing the mitigation strategies. The approach will accommodate differing perspective of uncertainties and risks associated with every country with respect to carbon emission to initiate a global trading system of carbon emissions.

Check Your Progress: 4

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. What are the broad areas in which the adaptation measures can be carried out?

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2. Give an account on CBDR.

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11.13 LET US SUM UP

The climate change is the biggest threat in the 21st century. The international scientific community urged the world to stabilize the level of emissions in the atmosphere in order to avoid future potential impact. In doing so, the world countries have proposed several international summits to discuss climate change and to find solution to the problems of climate change. This process is started in the year 1988 in Toronto as a World Conference on Changing Atmosphere. The politicians, scientists conclude in the Toronto Conference “Humanity is conducting an unintended, uncontrolled, globally pervasive experiment whose ultimate consequences could be second only to a global nuclear war”. The conference recommends reducing carbon dioxide emission by 20 %. In the same year Intergovernmental Panel on Climate Change (IPCC) met in Geneva and assessed the state of scientific knowledge on climate change, evaluated its impact and brought realistic solution. The first IPCC report was published in the year 1990. All these events led to the world community to conduct series of climate conferences. Through this unit, we discussed climate change as a global - political issue in the context of human’s contribution to climate change and the possibility of framing the action plan for combating the climate change events. Further, we have

studied about the Rio declaration on Environment and Development; significance of UNFCCC and IPCC; Kyoto Protocol and Paris Agreement on Climate Change.

11.14 KEYWORDS

Clean Development Mechanism : The CDM allows emission reduction(or emission removal) projects in developing countries to earn Certified Emission Reduction (CER) credits, each equivalent to one tonne of carbon dioxide. CERs can be traded and sold, and used by industrial countries to meet a path of their emission reduction targets under Kyoto protocol. (Source: <http://cdm.unfccc.int>)

South-South Cooperation: South-south cooperation is a new UNEP initiative that is being promoted as an essential cross cutting mechanism designed to enhance UNEPs ability to deliver capacity building and technology support activities in the field of environment in developing countries and regions of the south. The implementation of the south-south cooperation initiatives is being carried out in the context of Bali Strategic Plan (BSP) for technology support and capacity building. The BSP serves as an umbrella framework for coherent, coordinated and effective delivery of environmental capacity building and technical support activities in response to well defined countries priorities and need source:www.unep.org/southsouthcooperation).

Bonn Agreement: It is an informal term of political deal reached at COP-6 in Bonn, Germany in 2001 by which governments agreed on the most politically controversial issues under the Buenos Aires Plan of Action.

Human Security: Human security is an emerging paradigm for understanding global vulnerabilities whose proponents challenge the traditional notion of national security by arguing that the proper referent for security should be the individual rather than the state. Human security holds that a people- centered view of security is necessary for national, regional and global stability.

11.15 FURTHER READING

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11.16 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. The Berlin meeting had the following objectives:
 - To establish an adhoc committee to negotiate a protocol;
 - To initiate joint-activities among the participating countries to carry out a survey;
 - To use Global Environment Facility as the FCCC’s financial mechanism;
 - To locate FCCC permanent office at Bonn.
2. The primary principles included in the climate convention were as follows:

Target and Timetables: The European Union and AOSIS advocated to follow target and timetable approach to limit the emission level.

Financial Assistance and Transfer of Technology: The developing countries demanded for creating a new fund towards helping developing countries in implementing the Convention.

Institution and Implementation Mechanisms: OECD countries including United States advocated to establish implementation machineries including scientific body, deliberate meetings, reporting, and monitoring of the implementation work to tackle the climate change problem.

Check Your Progress: 2

1. The five key principles of Kyoto Protocol are:
 - Commitments
 - Implementation
 - Minimizing impacts
 - Accounting, reporting
 - Compliance

2. The major shortcomings of the Kyoto protocol are:
 - Excluding developing countries from the protocol. This will reduce the effectiveness of the agreement.
 - Exclusion of India and China which constitute largest population and growing economy.
 - Using of older, outdated and environmentally unfriendly technologies by the developing countries which are excluded from the Kyoto protocol.
 - Growing unchecked developing countries emission.
 - The Kyoto protocol only raises the awareness and not in real terms.

Check Your Progress: 3

1. The Bali action plan adopted by the COP includes long-term cooperative action which goes beyond 2012. The Bali action plan centered four main building blocks i.e. mitigation, adaptation, technology and financing. They also insisted the future discourse of action by all developed countries.
2. Copenhagen accord held in 2009 accepted the need to hold mean temperature increases below 2 degree Celsius and explicitly endorsed the dual track climate negotiations under the UNFCCC. Non annexure 1 countries also pledge to reduce emissions. It also establishes the principles for a system of international monitoring, reporting and verification. The accord paves the way for an increase in future funding for developing countries.

Check Your Progress: 4

1. The broad areas where the adaptation measures have been carried out include:
 - Crop Improvement
 - Drought Proofing
 - Health
 - Risk financing
 - Disasters management
 - Livelihood Preservation
2. The CBDR principle was derived from the principle of equity, which is an important principle of international law. For the first time, the Rio Declaration on Environment and Development was being instrumental in incorporating the principle of CBDR at the global level.

UNIT 12 INDIA'S POLICY ON CLIMATE CHANGE

Structure

- 12.0 Introduction
- 12.1 Learning Outcomes
- 12.2 Copenhagen Summit 2009
- 12.3 India and Copenhagen Summit
 - 12.3.1 India's Position
 - 12.3.2 India and BASIC Countries (Brazil, South Africa, India & China)
 - 12.3.3 Impact of India's Position on the Copenhagen Summit
- 12.4 India's Policy and Action towards Renewable Energy Sources
- 12.5 Paris Agreement
- 12.6 National Action Plan on Climate Change
- 12.7 Let Us Sum Up
- 12.8 Keywords
- 12.9 Further Readings
- 12.10 Check Your Progress: Possible Answers

12.0 INTRODUCTION

“We have the moral responsibility to bequeath to our children a world which is safe, clean and productive, a world which should continue to inspire the human imagination with the immensity of the blue ocean, the loftiness of snow covered mountains, the green expanse of extensive forests and the silver streams of ancient rivers” said the former Prime Minister of India, Dr. Manmohan Singh. In the last couple of decades, India has emerged as one of the fastest growing economies along with China and Vietnam (Basu, 2008). The structural shift that took place in Indian economy in the post 1980s has contributed in rising growth rate in Gross Domestic Product (GDP) from 5.4 per cent in 1980s to 6.4 in the 1990s and further increased to 8.4 percent per annum during 2003-04 and 2011-12 (Mehrotra, 2014). Government of India's adoption of Structural Adjustment Programme (SAP) in 1991 to bring macroeconomic stability (Kennedy, Loraine, 2014), trade liberalization and attract investment from multinational and transnational corporations have had a significant impact on the growth of the secondary and tertiary sectors, migration from rural to urban areas and reduction in the growth of primary sector. It also affected environment severely. At the same time, poverty in rural areas measured in consumption terms has substantially decreased to 26 percent in 2011-12 from 34 percent in 2009-10 and 42 percent in 2004-05 at all India level. Still India needs to go a long way. Developing countries like India need to accelerate economic growth to reduce poverty and improve socio-economic conditions. In this situation, it is important for the countries to consider equity principle.

Hope you remember global climate change negotiations like Rio, Kyoto and Bali Action Plan which you have learnt in the unit 13 of this course. Please recall the Kyoto Protocol negotiations. Before the Kyoto protocol expires at the end of 2012, both developing and developed countries of United Nations (UN) Climate Change Conference signatories aspired to strike another deal. In this backdrop, the Copenhagen conference was convened in December 2009. Before conference negotiations started, the developed world suggested that the developing countries like India and China must undertake emission reduction commitments. India has been assuring that its per capita emission level never exceed that of the developed world. The climate change management at the international level consists of 1992 United Nations Framework Convention on Climate Change (UNFCCC), 1997 Kyoto Protocol and the decisions taken by the signatory countries under these instruments. Even though, it was considered as a first step to address climate change and its consequences, it was not sufficient to force member States to take appropriate actions to address climate change with rapid change in environment and rising temperature. It was widely considered as an inadequate instrument to combat climate change. Both the UNFCCC and Kyoto Protocol did not contain necessary clauses to mitigate greenhouse gases (GHG) emissions, which resulted lack of binding obligation for developing countries. Countries like China took adverse steps and started emitting GHG like developed nations just to counter their behaviour. This fuelled further to the existing problems and some of the developing countries withdrew from Kyoto protocol in the second commitment period. Hence, Parties to the UNFCCC came together and formed an Ad-Hoc Working Group at the Durban Conference in 2011. It was called as Durban Platform for Enhanced Action (ADP) to negotiate a new climate agreement by 2015 that would come into force from 2020. The four years of continuous negotiating process resulted with the formulation of 2015 Paris Agreement which was expected to govern, regulate and incentivise the next generation of climate actions. With this introduction, through this unit, we will discuss Copenhagen Summit; Paris Agreement; National Action Plan on Climate Change; Greenhouse Gas inventory; and also a critical review of India's Action Plan and India's policy and action towards renewable energy sources.

12.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- describe Copenhagen summit 2009;
- explain India's position in Copenhagen Summit;
- analyse India's Action Plan in Copenhagen Summit;
- examine Paris Agreement and its relation with India; and
- describe National Action Plan on Climate Change.

12.2 COPENHAGEN SUMMIT 2009

The mandate of the 15th Conference of Parties (COP) in Copenhagen is to enhance long term cooperation on climate change under the Bali action plan.

It is not about renegotiating the United Nations Framework Convention on Climate Change (UNFCCC). The climate change attracted everyone in this world. The Intergovernmental Panel on Climate Change (IPCC) and the scientific community proposed number of actions and plans to halt global warming. The result is the climate change negotiation. The Copenhagen summit in 2009 was a great success. It has brought global environmental issue to the wider forum. Nearly 130 government delegates from 191 countries and over 40,000 scientists, activists and industrialists registered for the convention. Leaders who spoke in the convention quoted the scientific findings of the IPCC and they were aware about the threats of climate change and how it would affect people, food production, water supply and environment.

The Copenhagen summit was convened to negotiate post-Kyoto protocol agreement at the UN Framework Convention on Climate Change (UNFCCC) to sign a legally binding deal. The Bali action plan in the year 2007 laid a roadmap for the post-Kyoto Protocol agreement for developed countries (Annexure 1 parties in the convention) to reduce greenhouse gases (GHG) emission and assist developing countries to adapt climate change. The developed countries also agreed to help developing countries by transferring their technology to make them to adopt low carbon emission path way. The Bali climate convention has agreed two track negotiations: 1. “working group for long-term cooperative action”, 2. “working group on Kyoto protocol”.

IPCC set the targets for the Copenhagen negotiations. According to them, the GHG emissions will increase enormously by 2015 and then decline by 25-40% over 1990 level by 2020 and will reduce by 80 % over 1990 by 2050 to stabilize carbon dioxide concentration at 450ppm to limit the global warming below 2⁰C. (IPCC 2007). The analysis of World Resource institute shows that the commitments announced so far by rich countries may add up to only 13-19 % emission reduction whereas according to IPCC, 25-40 % reduction is needed (Levin and Bradley, 2009). According to the study by International Energy Agency, limiting global warming to 2⁰C with the carbon dioxide concentration limited to 450ppm is feasible, only if the world is ready to invest in low carbon technologies. The key issues discussed at Copenhagen summit are as follows:

- Making continuous progress in the negotiations of Kyoto protocol.
- Insisting governments to commit mid-term GHG emissions reduction.
- Developing scientific monitoring, reporting and verification methods.
- Funding for adaptation and mitigation.
- Transferring technology to the developing countries.

According to the assessment, even if the carbon dioxide concentration is stabilized at 450ppm which seems highly unlikely there is 26-78 % of risk of overshooting the 2⁰C goal (Meinshausen, 2005). The Copenhagen summit has taken into consideration of scientific communities view on restricting global warming below 2⁰C. After the debates and the intervention of Island countries and Least Developed Countries, the summit included 1.5⁰C as a target for future negotiations and considerations. This Copenhagen summit

also recognized the need to cooperate in achieving halting of the global and national emissions as soon as possible by keeping the special circumstances of developing countries and the need for economic development and poverty alleviation (Ravindranath, 2010).

Copenhagen green fund was established to support mitigation, adaptation, technology transfer and reducing emissions from deforestation. The accord also agreed to transfer technology by establishing a mechanism to do the same. The developed countries agreed to provide US\$ 30 billion immediately for the period of 2010-2012 and mobilize US\$ 100 billion per year for the developing countries.

Check Your Progress: 1

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. What are the issues discussed at Copenhagen Summit?

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2. What is known as Green Fund?

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12.3 INDIA AND COPENHAGEN SUMMIT

The Indian delegations in the Copenhagen Summit have expressed the positive outcome because it does not set the time limit for reducing the GHG emissions. India has to increase fossil fuel based energy to promote the economic growth. The assumption is that if it increases the fossil fuel energy, it automatically reduces the poverty.

India's expectation from Copenhagen Summit is as follows:

- International cooperation should be there to combat climate change.
- The outcome of the negotiations in the Copenhagen should be fair and equitable. It must be in accordance with the "principle of common but differentiated responsibilities and respective capabilities" as per 1992 Rio declaration.
- Summit should provide a space to accelerate socio-economic development in order to eradicate poverty through ecologically sustainable manner.

12.3.1 India's Position

1. India, as a sovereign State totally rejected any form of surveillance or supervision by international organisations in relation to CO₂ emission

reduction within the state, especially to the extent that:

- It did not want financial support from international agencies or organization. It was ready to manage from domestic resources.
- It was ready to take its own procedure for reduction of CO₂ emissions and the percentage reduction will remain at the discretion of the country.

2. The Copenhagen Conference is not a legally binding text. The Copenhagen Conference is only a procedure for improving the existing institutional framework to combat climate change and is not under any circumstances a re-negotiation of it nor can it result in the development of a new framework.

The UN Framework Convention on Climate Change adopted at the Rio de Janeiro Earth Summit, 1992 is the legally binding text. Based on this, the international community must operate in order to achieve its objectives and to deal with the phenomenon of climate change. It is fully applicable for India. The purpose of the Copenhagen Summit is to continue to implement, and to reinforce the existing institutional framework and not create a new one. In other words, the purpose of the Conference is to encourage the long-term collaboration under the Bali Action Plan and not renegotiate the UN Framework Convention on Climate Change (UNFCCC).

3. As a continuation of that position taken by India, the Kyoto Protocol remains the core, adequately binding text at international level, and consequently it was not considered necessary to prepare a new text. The Kyoto Protocol will remain in force even after 2012. The only subject for debate is determination of new restrictions on the quantity of CO₂ emissions for developed countries and the setting of a new time period for achieving those targets after 2010. The validity of the Kyoto Protocol cannot be allowed to cease after 2012 nor can the expectation for developed countries to undertake the obligation to limit the quantity of CO₂ emissions after 2012 be abandoned.
4. As we mentioned earlier, the developed economies are responsible for the phenomenon of underdevelopment in the global system. As a consequence, it is for certain that the responsibility of the developed countries to actively support sustainable development in developing countries by transferring economic resources and technology without enforcing conditions of any sort, such as patents or copyright. The parties responsible for the underdevelopment cannot set conditions.
5. It is unfair to impose emission norms to all industries at global level on issues relating to sectoral emission reduction policies. In other words, the types of industries which cause the greatest harm by emitting large quantities of CO₂ must be identified and focus should be given only on those industries.

12.3.2 India and BASIC Countries (Brazil, South Africa, India & China)

On the one hand, the developing countries like India and other countries in the BASIC group came together and took strong position on emission norms, and the developed countries on the other led by the USA withdrew from the Conference. The fragmentation of views and the closely-held views of various groups of States meant that the Conference had every prospect of failing.

The enabling conditions have emerged as a result of multilateral pressures and negotiations during the Copenhagen Conference, mainly from the USA, coupled with the personal attendance of former US President Obama at the Conference. This would not have been the case if, in the meantime, the necessary consent had not been obtained from the large number of States. These States have been created as number of blocs. Finally, the Copenhagen Accord successfully emerged (<http://unfccc.int/resource/docs/2009>).

India has announced a voluntary reduction of CO₂ emissions by 20-25% up to 2020 compared to a 2005 baseline for emissions, particularly through a process of implementing policy measures which included the mandatory application of quality characteristics to fuels. Both the USA, and China, as well as India, have already submitted plans to curtail emissions. More specifically, on 8.3.2010 the Government of India's Minister for Environment & Forests in a letter to the Executive Director of the UNFCCC Secretariat, Mr. Yvo de Boer, pointed out that "Please refer to your letter vide no. YdB/DBO/JBU/eps/log286-10 dated 3 February, 2010 regarding listing in the chapter of the Copenhagen Accord. It may be recalled that India actively participated in the discussions on the Copenhagen Accord. India stands by the contents of the Accord. Our clear understanding is that the Accord is a political document. It is not legally binding. The Accord is meant to facilitate the ongoing negotiations in the two tracks in accordance with the principles and provisions of the UNFCCC, the Kyoto Protocol and the Bali Action Plan. The Accord was not adopted by the Conference of Parties but just taken note of. However, the Accord could have value if the areas of convergence reflected in the Accord are used to help the Parties reach agreed outcomes under the UN multilateral negotiations in the two tracks, i.e., the Ad-hoc Working Group on Long Term Cooperative Action and the Ad-hoc Working Group on Kyoto Protocol. The Accord is only an input into the two-track negotiations. The Accord is not /a new track of negotiations or a template for outcomes." (www.usclimatenetwork.org/policy/copenhagen-accord-commitments#note10).

Consequently, the position of the Indian Government which emerged as a result of the negotiations which took place during the Conference is, to a greater or lesser extent, different from the original ones.

12.3.3 Impact of India's Position on the Copenhagen Summit

1. India has emerged as regional leader in the SAARC region by strengthening its economy.

2. Development of basic block in the conference is the result of India's aspiration as a regional leader to foster adhoc partnership.
3. It has strengthened its relationship with China.
4. The closer ties of India and China on climate change issues are expected to bring multiple impact at the international level.
5. India strengthened its position by mobilizing more developing countries to adopt joint position and shape the final conference text. So the text was not binding. It acted as a powerful member for the economically developing force.
6. The role played by the G77 was worth to be noted and India being part of G77 group had a significant influence.
7. The conference proceedings showed that the developed countries have splitted into many and least developed countries took different positions. The island nations had a different position. This affected the outcome of the conference proceedings and the conference could not take specific time frame for implementing the climate change.

Check Your Progress: 2

Note: 1) Use the space given below for your answers.
2) Check your answers with those given at the end of this unit.

1. List out the expectations of India from Copenhagen Summit.

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12.4 INDIA'S POLICY AND ACTION TOWARDS RENEWABLE ENERGY SOURCES

This section is the official announcement of the Government of India for the post Copenhagen domestic actions. The following are some of the policies framed and implemented by India towards renewable energy sources (Ministry of Environment and Forest, 2010).

1. Expert group was set up on low carbon strategy for inclusive growth

The representation from industry, leading think tank, research institutions and civil society and government are part of this expert group. The group has given mandate to develop a road map for India for low carbon development. It will recommend prioritized actions in sectors such as electricity, transport, etc.

2. A "carbon tax" on coal to fund clean energy

India has announced a levy – a clean energy cess – on coal, at the rate of Rs. 50 (~USD 1) per ton, which will apply to both domestically produced and imported coal.

- This money will go into a National Clean Energy Fund that will be used for funding research, innovative projects in clean energy

technologies, and environmental remedial programmes.

- The expected earnings from this cess is around USD 500 million for the financial year 2010-11.

3. **Perform, Achieve and Trade (PAT) mechanism for energy efficiency**

India's cabinet approved the National Mission on Enhanced Energy Efficiency (NMEEE) on 24th June, 2010. The Mission includes several new initiatives – the most important being the Perform, Achieve and Trade (PAT) mechanism, which will cover facilities that account for more than 50% of the fossil fuel used in India, and help reduce CO₂ emissions by 25 million tons per year by 2014-15.

- a. About 700 of the most energy intensive industrial units and power stations in India would be mandated to reduce their energy consumption.
- b. In order to enhance the cost effectiveness of this mechanism, facilities which achieve savings in excess of their mandated reduction would be issued Energy Savings Certificate (ESCerts) for the savings that are in excess of their mandated target. These ESCerts can be used by other facilities for compliance if they find it expensive to meet their own reduction target.
- c. Energy efficiency ratings made mandatory for 4 key appliances—refrigerators, air conditioners, tube lights and transformers.

4. **Release of India's National GHG inventory**

Parties to the United Nations Framework Convention on Climate Change (UNFCCC) required to submit periodical report on emissions according to the paragraphs 8-24 of the annex to decision 17/CP.8 meant for reporting of National Communications (NC) from Non-Annex I parties. India being a signatory of UNFCCC prepares national inventory on emissions by collecting quantitative data. The data will be collated, analysed and reported in the form of table.

India released its Greenhouse Gas (GHG) Emissions Inventory for 2007, with the aim of enabling informed decision-making and to ensure transparency.

- With this publication, India has become the first “non-Annex I” (i.e. developing) country to publish such updated numbers.
- India also announced its intent to publish its emissions inventory in a two-year cycle going forward, which is much more frequent than the requirement under its NATCOM commitments. India will be the first developing country to do so.
- According to the results, India's emissions are less than a fourth of the USA and China.
- Results also show that the emissions intensity of India's GDP declined by more than 30% during the period 1994-2007 due to the efforts and policies that India has proactively put in place.
- Despite its already low emissions intensity, India intends to do even more. India has announced its intent to further reduce the emissions

intensity of its GDP by 20-25% between 2005 and 2020, even as it accelerates infrastructure development and the growth of its manufacturing sector.

We have given GHG emissions by sector wise for 2014 in the table 12.1

Table 12.1: Greenhouse Gas Emissions by sectors in India by 2014 (Gg)

	CO ₂ emission	CO ₂ removal	CH ₄	N ₂ O	HFC23	CF ₄	C ₂ F ₆	SF ₆	CO ₂ equivalent
Total without LULUCF (Gg)	1,997,891.85	-	20,005.35	475.29	1.59	2.61	0.71	0.004	2,607,488.12
Total with LULUCF (Gg)	2,015,107.03	319,860.23	20,053.54	476.71	1.59	2.61	0.71	0.004	2,306,295.43
1. Energy	1,844,705.03	-	2,133.37	65.35	-	-	-	-	1,909,765.74
2. IPPU	153,186.81	-	177.85	10.36	1.59	2.61	0.71	0.004	202,277.69
3. Agriculture	-	-	14,709.78	349.39	-	-	-	-	417,217.54
4. LULUCF	17,216.04	319,860.23	48.19	1.42	-	-	-	-	-301,192.69
5. Waste	-	-	2,984.35	50.18	-	-	-	-	78,227.15
Memo Item (not accounted in total Emissions)	812,030.60	-	0.11	0.11	-	-	-	-	812,067.87
International Bunkers	4,943	-	0.11	0.11	-	-	-	-	4,980.811
Aviation	3,681.65	-	0.03	0.10	-	-	-	-	3,714.12
Marine	1,261.88	-	0.08	0.01	-	-	-	-	1,266.69
CO ₂ from Biomass	807,087.06	-	-	-	-	-	-	-	807,087.06

IPPU: The Industrial Processes and Product Use

Source: MoEFCC. (2018). India: Second Biennial Update Report to the United Nations Framework Convention on Climate Change. Ministry of Environment, Forest and Climate Change, Government of India.

As per the table 12.1, India's emissions in 2014 were 2,607,488.12 Gg of CO₂e of greenhouse gases (GHGs) without Land Use, Land-Use Change and Forestry (LULUCF). The following are the excerpts from the report.

“The energy sector accounted for 73% of the total GHG emissions for the year 2014. Fuel combustion activities emitted 1,871,709 Gg CO₂e in 2014 including 1,140,983 Gg CO₂e from energy industries. Within energy industries, 94.96% of emissions were from electricity production, 4.39% from refinery and 0.66% from manufacturing of solid fuels. Thus, electricity production accounted for about 42% of the entire GHG emissions from all the sectors in 2014. The manufacturing industries and construction together emitted 351,909.54 Gg CO₂e, which was 18.4% of total emissions from the energy sector. The sub-categories that share the total contribution to the emissions from the manufacturing industries are: Cement (13.4%), Iron & steel (43.9%), Non-ferrous metals (0.5%), Chemicals (0.6%), Pulp & paper (1.1%), Mining & quarrying (0.9%), Textile/leather (1.0%), Bricks (0.8%), Fertilizer (1.7%), Engineering Sector (0.1%), and Non- specific Industries (36%)”.

The report further mentioned that “Road transport accounted for 90.1% of the total emissions from the transport sector, followed by civil aviation (5.6%),

railways (3.1%) and domestic water-borne navigation (1.2%). In 2014, other sectors in the energy sector together emitted 128,643 Gg of CO₂e, of which approximately two-thirds were contributed by the residential sector, about one fifth by the commercial sector and rest by the biomass burnt for energy (non- CO₂ GHGs) and the agriculture/fisheries sectors put together. A comparison of fossil fuel combustion emissions using both, the reference approach and the sectoral approach was also conducted, and the difference was found to be 3.83%. The total fugitive emissions in the year 2014 were 38,057 Gg CO₂e, of which 43% was from coal mining and post mining operations in India. Fugitive methane emissions have registered a decrease of 22% between 2010 and 2014, mainly due to a relative reduction in underground mining activities. Fugitive emissions contributed to 2% of emissions from the energy sector”.

“The Industrial Processes and Product Use (IPPU) category emitted 202,278 Gg of CO₂e in the year 2014, and accounted for 8% of the total GHG emissions. Within IPPU, cement production is the largest emission source, accounting for about 57% of total IPPU sector emissions. In total 42,145 Gg CO₂e was emitted as fluorinated gases in 2014. Agriculture sector is the main source of methane (CH₄) and nitrous Oxide (N₂O) emissions. CH₄ emissions occur from this sector mainly due to livestock rearing (enteric fermentation and manure management) and rice cultivation. N₂O is principally emitted due to the application of fertilizers to the agricultural soils. In the year 2014, the agriculture sector emitted 417,218 Gg of CO₂e, which amounted to around 16% of the emissions of India for that year. Of these, 74% was CH₄, and 26% was N₂O. LULUCF sector was a net sink of 301,193 Gg CO₂e during 2014, registering an increase in the sink activity of the sector. Cropland dominates the CO₂ emissions /removal estimates for India for the year 2014. Forest land, cropland and settlement categories were net sinks while grassland was a net source of CO₂. About 12% of India's GHG emissions were offset by the LULUCF sector. The waste sector contributed to 3% to total GHG emissions in 2014. The waste sector was dominated by emissions from wastewater handling which account for more than 80% of the sectoral emissions. Methane from solid waste disposal was 717 Gg whereas CH₄ emissions from wastewater treatment and discharge, including domestic, commercial and industrial wastewater were 2,267 Gg. Domestic and commercial wastewater handling also emitted 50 Gg of N₂O.”

5. National Mission on Sustainable Habitat

The NMSH was recently approved as one of the eight National Missions under the Prime Minister's National Action Plan on Climate Change (NAPCC). A comprehensive strategic plan is being drafted for the implementation of this mission.

6. Jawaharlal Nehru National Solar Mission

The JNNSM which has been launched recently is an ambitious mission to make India a global leader in solar energy. This is also one of the eight National Missions under NAPCC.

- The National Solar Mission aims at generating 20,000 MW of solar power by 2022.

- The Mission also has other targets: 2000 mw of off-grid solar plants, and 20 million square meters of solar collectors to be installed. In addition, 20 million solar lighting systems will be created/distributed in rural areas, saving about 1 billion litres of Kerosene every year.

7. Green India Mission

The GIM, also one of the eight National Missions under NAPCC, is being finalised. The overarching target of the GIM is to double the area to be taken up for afforestation/eco-restoration in India in the next 10 years, taking the total area to be afforested or eco-restored to 20 million ha. This would increase the above and below ground biomass in 10 million ha of forests/ ecosystems, resulting in increased carbon sequestration of 43 million tons CO₂e annually.

Key features of GIM include:

- Increasing the quality of our forest cover by increasing the cover and density of our medium density and degraded forests.
- Taking a holistic view of forestry, and not merely focus on plantations to meet carbon sequestration targets.
- Focusing on decentralization and involving existing local governance institutions. Forests are the main source of livelihood to over 200 million people in India and hence GIM will actively try to secure the participation of local communities.

8. Regional and International Cooperation

- South Asian Association for Regional Cooperation (SAARC), which comprises the 8 South Asian countries, adopted the Thimpu Statement on Climate Change on 29th April 2010. The Statement calls for, among other things, an Intergovernmental Expert Group on Climate Change to develop clear policy direction for regional cooperation on climate change.
- India announced the grant of USD 1 million each to SAARC Forestry Centre, Thimpu, Bhutan and SAARC Coastal Management Centre, Male, Maldives.
- India and Bangladesh to set up the India-Bangladesh Sunderbans Ecosystem Forum to conserve the Sunderbans – the world's largest riverine delta. Climate change will be the central component of this Forum.

9. Climate Change Science

- The Indian Network for Climate Change Assessment (INCCA) is undertaking a major “4×4” assessment of the impacts of climate change on four sectors – water resources, agriculture, forests and human health – in four critical regions of India – the Himalayan region, North east, Western Ghats and Coastal India.
- INCCA is basically a network consisting of 127 research organizations. Its task is to perform research in the domain of climate change and climate change impacts on various sectors of

10. India's first CDM PoA (Bachat Lamp Yojana)

The Bachat Lamp Yojana (BLY) conceived as CDM Programme of Activity (PoA) for mass distribution of Compact Fluorescent Lamps (CFLs) in India has been registered successfully by the CDM-Executive Board.

11. Himalayan Ecosystem

The mission aims to evolve policy and management strategies for protecting and sustaining the Himalayan mountain ecosystem including Himalayan glaciers. Further, it aims to establish “an observational and monitoring network” to assess the resources and ecosystem health of Himalayan ecosystem.

12.5 PARIS AGREEMENT

The successful multilateral diplomacy has brought a historic Paris Agreement to address climate change. The Paris Agreement is a ‘treaty’ under the Vienna Convention on the Law of Treaties, 1980. It legally binds all States that have expressed their consent to be bound by this agreement. Once they accept and approve, they ratify the same and take necessary measures to implement the same by formulating laws, policies and programmes.

Any legal character of a provision in the international treaties makes the parties to oblige legally. It also set the standards for States to achieve the obligations within prescribed time period. States need to assess their work in a regular interval and submit reports periodically. This may give clear picture whether states have compliance/non-compliance to the agreed provisions. If we assess Paris Agreement through these lenses, states need to fully oblige mitigation and transparency sections of the Agreement. Unlike mitigation and transparency sections, adaptation sections seem to be soft obligations.

175 countries including India ratified Paris Agreement. As we mentioned earlier, unlike the divisive Kyoto Protocol, all parties required to submit relevant documents with regard to GHG mitigation and adaptation measures to counter adverse effects of climate change as per Paris Agreement at regular intervals. 189 states/countries covering over 95% of global emissions have submitted nationally determined contributions in the context of Paris Agreement. The Paris Agreement has aspired to address adverse effects of climate change by stabilizing temperature rise (‘well below 2°C’ and to aspire to 1.5°C). As we mentioned earlier, each state/country must report to their efforts to combat GHG emissions or mitigation measures in every five years.

Agreement provides autonomy to the states/countries to decide on their own to address climate change. But, their successive efforts must be progressive one compared to previous ones. There are three important components in the Paris agreement. Firstly, states/countries must take transparent measures to address or mitigate climate change and their agreed contribution. Secondly, there will be global stock taking process which will be assessed periodically

to recognize collective progress towards the Agreement's long term goals. Thirdly, there will be a compliance system that facilitates member states/countries. It was recognized by the previous studies that the measures taken to mitigate and adapt to climate change might affect human rights. First of its kind, Paris Agreement accepted the relationship between climate change and human rights like right to life, right to health, right to food, and right to housing, with marked departure from earlier climate change instruments. It contained explicit reference. In addition to this, states/countries need to formulate necessary laws to improve socio-economic conditions of vulnerable to address human rights issues. Yet, until recently, no legally binding international climate instrument explicitly recognised the existence of intersections between human rights concerns and climate change.

As far as India is concerned, the Paris Agreement requires India to submit its national contribution every five years, ensuring that each contribution is a progression on the previous one. It also requires India to report periodically on its actions to achieve and implement its contribution. In addition, India should 'strive' to submit long-term low-GHG development strategies within which these national contributions will sit. The Paris Agreement makes India to formulate necessary policies to address energy requirements, environment and development. As mentioned earlier, each member states need to submit periodically about their progress. To do the same, India must collect quantitative data rigorously to show the progress. Qualitative data may be helpful to recognize the impact of policies on vulnerable. Although, India was able to reduce poverty among vulnerable considerably over plan period, it needed to take further measures to address poverty and vulnerability in certain geographical areas. The measure taken at this issue must be in a right direction. India has responsibility at the international level too. With regard to reducing GHG emissions to bring temperature at certain levels, countries must share burden equitably. Countries like India must give voice in this direction.

12.6 NATIONAL ACTION PLAN ON CLIMATE CHANGE

National Action Plan for Climate Change (NAPCC) was formulated and released on 30th June, 2008. The guiding principles for NAPCC are given below:

- Formulation of inclusive, climate change policies by protecting the rights of the poor and vulnerable;
- Achieving national growth objectives by taking measures in a definite direction that must address ecological sustainability which further address greenhouse gas mitigation;
- Devising efficient and cost effective strategies for end use demand side management;
- Developing and deploying appropriate technologies for both adaptation and mitigation of greenhouse gases emission extensively at an

accelerated pace;

- Encouraging all stakeholders to develop a new and innovative forms of market, regulatory and voluntary mechanisms to promote sustainable development;
- Ensuring effective implementation of programme through unique linkages, including with civil societies and local government institutions and through public-private partnership;
- Encouraging international cooperation for research, development, sharing and transfer of technologies enabled by additional funding and a global IPR regime that facilitates technology transfer to developing countries under UNFCCC.

Based on the above mentioned guiding principles, the Government of India has developed eight National Missions.

1. National Solar Mission (NSM);
2. National Mission for enhanced energy efficiency (NMEEE);
3. National Mission on Sustainable Habitat (MNSH);
4. National Water Mission (NWM);
5. National Mission for sustaining the Himalayan Ecosystem (NMSHE);
6. National Mission for a Green India (GIM);
7. National Mission for sustainable agriculture (NMSA); and
8. National mission on strategic knowledge for climate change (NMSKCC).

These missions will be implemented through different ministries based on their subject matter. Apart from ministries assigned to the task, Ministry of Finance, Planning Commission, experts from industry, academics and civil society will be part of the mission. The institutional structure depends upon task allocated for particular mission. Each ministry implement the mission by formulating necessary policies and programmes and develop detailed mission document which will be placed in the individual ministry website. We will discuss mission goals, strengths and weaknesses.

National Solar Mission

Mission has ambitiously targeted to generate considerable percentage of solar power by each state in India individually and country as a whole. The Mission's Renewable Purchase Obligations (RPOs) and auction process has brought fair degree of transparency and accountability to the process. According to the reports, the mission could able to achieve a substantial amount of work in the Phase I. The Mission has failed to recognize the potential off-grid generation which might helpful for poor and vulnerable. India is implementing various sustainable development programmes from first five year plan period onwards. The success of solar water heater scheme of Ministry of Non-Conventional and Renewable Energy in 1990s in the State of Karnataka and Maharashtra might be notable example for emulation.

National Mission for Enhanced Energy Efficiency (NMEEE)

The mission has introduced innovative measures like Perform, Achieve and Trade (PAT), Super-Efficient Equipment Programme (SEEP), and Market Transformation for Energy Efficiency (MTEE). The mission has made available necessary funds to achieve targets like Partial Risk Guarantee Fund (PRGF) and Venture Capital Fund for Energy Efficiency (VCFEE). The mission target was not holistic rather it targets only sub-sectors. It concentrates only on large scale industries.

National Mission on Sustainable Habitat (MNSH)

Mission plan was integrated with already existing plan Jawaharlal Nehru Urban Renewal Mission and tried to bring smart cities with energy efficient components. Mission must integrate their ideas by using bottom up approach to the planning and all its planning should be from stakeholders' perspective.

National Water Mission (NWM)

Mission affirms to bring comprehensive data base on the existing water sources. Study must be conducted on the relationship between water and climate change. Identifying existing water sources and restoring and rejuvenating with the help of communities are significant.

National Mission for Sustaining the Himalayan Ecosystem (NMSHE)

The mission recognises the importance of the Himalayas in sustaining large number of people and serves as an opportunity to promote a mountain-driven rather than plains-driven approach. But, the mission inadequately recognizes the significance of research to understand entire Himalayan ecosystem. Poor and vulnerable people are directly affected by the deforestation, depletion of natural resources and the construction of dams.

National Mission for a Green India (GIM)

Mission has ambitious plan to regulate and institutionalize local community level institutions. Mission will identify best practices and same will be used as learning sites. It failed to address the issues of reducing forest density.

National Mission for Sustainable Agriculture (NMSA)

Mission understood the problems faced by country especially agriculture sector with regard to climate change. But mission failed to understand the problems of marginal and small farmers. Identifying and scaling up of agriculture based on agro-climatic zone is important. The mission fails to address fuel and fertiliser shortages. In the era of post-globalization and mechanization of agriculture, the sector altogether faces different kinds of problem.

National Mission on Strategic Knowledge for Climate Change (NMSKCC)

It is significant to encourage all stakeholders to do research on long-term consequences of climate change. Knowledge gap and the impact of socio-economic changes must identified through research.

Check Your Progress: 3

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of this unit.

1. What is GIM? Explain its key features.

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2. What are the eight missions under National Action Plan on Climate Change (NAPCC)?

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12.7 LET US SUM UP

Through this unit, we have discussed that India has taken a firm and reasonable stand towards climate change negotiations. At international level, it mobilized likeminded countries to take appropriate decisions with regard to climate change. You can see that India is firmly committed for sustainable development and at the same time not compromising with its economic growth to reduce poverty. The final section of this unit has shown the measures taken by India towards renewable energy sources.

12.8 KEYWORDS

Annexure 1 Countries: Annexure 1 countries are industrialised countries and economies in transition.

LCA: Life Cycle Assessment (LCA) is the broadest indicator under internationally standardized method to evaluate the impact of climate change. It also assess acidification potential, eutrophication potential, ozone depletion potential and ground level ozone creation.

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12.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. The key issues discussed at Copenhagen summit are as follows:
 - a. Continuing negotiations of Kyoto protocol.
 - b. Making governments to commit midterm GHG emissions reduction.
 - c. Developing, monitoring, reporting and verification methods.
 - d. Funding for adaptation and mitigation.
 - e. Transferring technology to the developing countries.

2. In Copenhagen green fund was established to support mitigation, adaptation, technology transfer and reducing emissions from deforestation. The accord also agreed to transfer technology by establishing a mechanism to transfer.

Check Your Progress: 2

1. India's expectation from Copenhagen summit
 - a. International cooperation should be there to combat climate change.
 - b. The outcome of the negotiations in the Copenhagen should be fair and equitable. It must be in accordance with the principle of common but differentiated responsibilities and respective capabilities as per 1992 Rio declaration.
 - c. Summit should provide a space to accelerate socio-economic development in order to eradicate poverty through ecologically sustainable manner.

Check Your Progress: 3

1. GIM means Green India Mission. Key features of GIM include:
 - a. Increasing the quality of our forest cover by increasing the cover and density of our medium density and degraded forests.
 - b. Taking a holistic view of forestry, and not merely focus on plantations to meet carbon sequestration targets.
 - c. Focusing on decentralization and involving existing local governance institutions. Forests are the main source of livelihood to over 200 million people in India and hence GIM will actively try to secure the participation of local communities.
2. The eight National Missions under National Action Plan on Climate Change are as follows:
 1. National Solar Mission (NSM);
 2. National Mission for enhanced energy efficiency (NMEEE);
 3. National Mission on Sustainable Habitat (MNSH);
 4. National Water Mission (NWM);
 5. National Mission for sustaining the Himalayan Ecosystem (NMSHE);
 6. National Mission for a Green India (GIM);
 7. National Mission for sustainable agriculture (NMSA); and
 8. National mission on strategic knowledge for climate change (NMSKCC).

UNIT 13 STATE LEVEL ACTION PLAN

Structure

- 13.0 Introduction
- 13.1 Learning Outcomes
- 13.2 Policy Formulation
- 13.3 Agencies Involved in Policy Formulation in India
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13.0 INTRODUCTION

India is one of the fastest growing economies of the world with deficient energy sources to meet growing economy. India has more than one billion population. Out of them, 800 million people (79.9 percent of the population) live on less than US \$ 2 per day. More than 700 million people cook on traditional cook stoves using crop waste and animal residue. More than 400 million people do not have access to electricity. India stands at 128th position in the World Human Development Index. In the unit 14 of this course, we read International Conventions and India's response to the same. In response to the International Conventions, India has taken number of efforts to mitigate the impacts of climate change on various sectors. One such measure is National Action Plan on Climate Change (NAPCC). This was discussed in detail in the previous unit. NAPCC was part of the policy formulation. It is significant for the learners to understand policy formulation process before reading the efforts of the State Governments in India. This unit specifically talks about the efforts of the State governments to address climate change. We have taken few State governments as case examples. We are witnessing the effects of climate change by observing the increase of temperature, rise of sea levels with a perceivable increase in severity and frequency of extreme events (IPCC 2007a; SREX, 2012) and the speed of the change are seemingly fast. This led to a complex situation, whereby all spheres of life is getting impacted. Various studies related to greenhouse gases (GHG) emissions projected that the situation will escalate further, if GHG concentrations in the atmosphere from anthropogenic sources is not curtailed. The Governments at the National and States levels have taken notable policy decisions to address climate change related issues in various sectors. Before reading the efforts of

few state governments to address and mitigate climate change in the form of case studies, let's learn about policy formulation process.

13.1 LEARNING OUTCOMES

After studying this unit, you should be able to:

- discuss the process of policy formulation;
- explain the role of various agencies in policy formulation; and
- examine the efforts of state governments in India to address climate change related issues.

13.2 POLICY FORMULATION

As you recognize, there are various stages in the policy formulation process. The first step in policy formulation is identifying issues. Let's see what is Policy formulation? Hill (1993: p.47) defines 'policy' as 'the product of political influence, determining and setting limits to what the state does'. To explain further on this statement, it is the efforts of the governments to address the social problems by addressing issues scientifically to solve the same. Thus, it is referred as public policy (Anderson 1975). According to the political scientists', policy formulation is a process rather than one time affair. There are various stakeholders involved in public policy formulation process like political parties, civil society organizations, individuals, elected representatives, local bodies apart from executives and representatives of legislatures. Thus, the process of policy formulation includes drafting, consultation, negotiation, reformulation, finalization, legislation, converting policies in to programmes to achieve desired results. The success of any public policy depends upon identifying issues, drafting policy, and making all stakeholders to involve in the process actively and converting the policy into programmes for implementation.

“We need to ask number of questions when we take up policy formulation. According to Cochran and Malone, policy formulation takes up the “what” questions: “What is the plan for dealing with the problem? What are the goals and priorities? What options are available to achieve those goals? What are the costs and benefits of each of the options? What externalities, positive or negative, are associated with each alternative?”

As Dye puts it, the first stage in the policy making process, i.e. policy formulation which takes place anywhere mentioned in the following places depends upon the interest of the individual or groups. For example, there are chances to formulate policies in the government bureaucracies, interest group offices, legislative committee rooms, meetings of special commissions, or think tanks with details often formulated by those groups of people. Thus, taking the earlier mentioned aspects of policy formulation regarding climate change, Government needs to formulate relevant policies to address the impact of climate change, mitigation measures, sensitizing people to identify solutions locally, involving every tiers of government to address issues as part of the policy formulation process. Thus, any policy related to climate change will address the impact of the same.

13.3 AGENCIES INVOLVED IN POLICY FORMULATION IN INDIA

There are various agencies involved in policy formulation. India adopted parliamentary system of government with quasi federal structure. The formal way of policy formulation process is through Parliament and State Assemblies. Both executives and political wings of the government involve in the process. Apart from that, there are large of number of agencies/organisations involve in the process of policy formulation directly/indirectly. Informally, civil society groups, individuals, various associations, non-governmental organizations involve in policy formulation in the way of providing suggestions to improve policies. Cooperation and interaction among various players through formal and informal channels provide the basis for a more informed policy dialogue.

The policy formulation process in India is shaped within the framework of the constitutional system. The policy should reflect the will of the people. As far as policy formulation is concerned, the role of the administrative executive is to assist the ministers. The ministers are responsible and answerable to the parliament and legislature, regarding policy matters. A conscious effort has been made to define and demarcate clearly the areas of policy formulation between the central and state governments by listing the subjects in the central list, state list and concurrent list. Usually the central government enjoys great powers for the formulation of policies in the development sectors of the economy. However, implementation of policies involve both the central and state governments and the two have jointly to share responsibilities for results.

The enactment of 73rd and 74th Constitutional Amendment Act paved the way for delegating certain responsibilities to the local governments (both Urban and rural local bodies) in certain specific subjects related to the development of local areas. The notified subjects by the respective State governments to the local bodies in India are listed in the article 243G of Eleventh Schedule. The local bodies are free to implement policies and make necessary changes in the existing policies to suit the needs of their areas. The success of the policy depends upon the successful implementation of the same by the local governments whether it is solid waste management or successful maintenance of common properties or recharge of water bodies or maintenance of water bodies or involving local people to address climate change and mitigation measures.

In India, Parliament and State Assemblies serve as a forum for public debate on the issues of public policies. Apart from legislative bodies, National Development Council and Niti Aayog are also part of public policy formulation. They involve directly in the process of policy formulation. In order to facilitate the process of policy formulation, 'Policy Planning Units' are set up in key departments and ministries. They serve as effective policy making units for various programmes by giving scientific inputs and draft the policies. The success of any public policy depends on various factors like funding, administrative difficulties for implementation, repercussions likely

to emerge from the policy, methods of dealing with policy problems, etc. The table 13.1 provides a glimpse of public policy formulation on climate change.

Table 13.1: Ministries/Experts' Groups responsible for Climate Change Action in India

Cross Sectoral	Description of Participating Ministries
Prime Minister's Council on Climate Change	Prime Minister (Head of Council); Minister for External Affairs; Union Finance Minister; Union Minister for Environment, Forests and Climate Change; Union Minister for Water Resources, River Development and Ganga Rejuvenation; Union Minister for Agriculture and Farmers' Welfare; Union Minister for Urban Development; Union Minister for Science and Technology; Union MoS of Power, Coal and NRE; Principal Secretary to PM (Member-convenor of this panel); Cabinet Secretary; Foreign Secretary; Secretary, Ministry of Environment, Forests and Climate Change
Expert Group on Low Carbon Strategies for Inclusive Growth	Expert advisory group with domain experts from different areas/ sectors
Ministry of Environment, Forest and Climate Change	Nodal
Ministry of Finance	Cross-sectoral
Sector Specific	
Energy Sector	Ministry of Power; Bureau of Energy Efficiency; Central Electricity Authority; Central Electricity Regulatory Commission; State Electricity Regulatory Commission; Ministry of Power; Ministry of Coal; Ministry of New and Renewable Energy.
Transport	Ministry of Urban Development; Ministry of Road Transport and Highways; Central Pollution Control Board; State Pollution Control Board.
Industry	Ministry of Industry; Ministry of Iron and Steel; Ministry of Power; Ministry of Coal; Ministry of New and Renewable Energy; Bureau of Energy Efficiency.
Agriculture	Ministry of Agriculture and Farmers' Welfare; DARE
Buildings and Waste	Ministry of Urban Development; Ministry of Housing and Urban Poverty and Alleviation.
Forest	Ministry of Environment, Forest and Climate Change.
Strategic Knowledge Management on Climate Change	Ministry of Science and Technology

Source: Agarwal, Swati and Neha Pahuja. (2015). Green Growth and Climate Change Mitigation in India. New Delhi: TERI.

Check Your Progress: 1

Note: 1) Use the space given below for your answers.
2) Check your answers with those given at the end of the unit.

1. Define policy.

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2. Give a brief account of policy formulation.

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13.4 STATE GOVERNMENTS' EFFORTS TO ADDRESS CLIMATE CHANGE: STATE ACTION PLAN

As we mentioned earlier in this unit, Constitution of India clearly demarcated the roles and responsibilities of Centre, State and Local Governments. Government of India is responsible to make necessary laws on subjects listed in the “Union List” and “Concurrent List”. Thus, Central government formulated NAPCC and requested all State governments to formulate State Action plan on Climate change (SAPCC). In 2009, the Government of India notified state governments to come up with their own climate change policies related to mitigation and adaptation. The plans and actions should be aligned with their respective state priorities as well as the NAPCC goals. Hence, the State Action Plan on Climate Change (SAPCC) in India is the most notable policy reflecting the nature of decentralized climate change mitigation framework of the country.

The State governments need to take following steps before implementing “State Action Plan on Climate Change”. It required to seek prior approval from the Ministry of Environment, Forest and Climate Change (MoEFCC) before implementation of activities listed therein. For implementation of SAPCCs, a combined budgetary requirement of INR 11.32 lakh crores (USD 188.66 billion) has been received by the ministry (MoEFCC, 2014). Ministry will scrutinize the proposal and disburse the required amount as per needs of the State governments. However, States may receive all necessary help from the centre to implement their plans to achieve national targets and priorities related to climate change.

India has vast geographical area having varied agro-climatic conditions. It is important to formulate policies by considering all these factors. Central government broadly frame the policies and state and local governments make necessary changes in the same to suit their local conditions and implement the same within the decentralized framework of mitigation strategies. India as mentioned earlier is a federal State and always involves every level of

governance structure for a critical policy intervention. Delivery of services and implementation will be effective only if all are involved in it. With regard to climate change policy, disaggregated climate data clearly indicates that the vulnerabilities and opportunities will be well understood at the state and local level than at the country level. Through states, national policies reach to a large number of sectors and actors effectively while addressing state priorities at the same time. In addition, measurement and monitoring at the state level can be taken up more efficiently. For this reason, many of the national policies in India are implemented through the states. Some other standalone programmes like renewable purchase obligations, implementation of feed-in-tariffs, setting up of state energy conservation funds, implementation of small and hybrid hydroelectric systems and energy conservation building codes are some of the examples of policies that are implemented through the states. Almost all of these now come together within the larger umbrella of the state action plans.

It is significant for the learners to remember the following points related to policies.

- The Ministry of Environment, Forest and Climate Change requested all Indian States to develop action plans by defining their activities and programmes to be undertaken with the aim of adaptation and mitigation in consonance with the objectives of NAPCC.
- The objectives of the NAPCC were to ensure sustainable development in the country. It will provide guidance to formulate necessary policies in a sustainable manner in future. This can be achieved only through the active support of State and local governments. Hence, the Government of India has encouraged the State Governments to develop State Action Plan on Climate Change (SAPCC).
- SAPCCs would address the need of the country as well as provide direction to achieve the same. It would help identify measures that promote our development objectives while also yielding co-benefits for addressing climate change effectively.
- It wants to advance India's development and climate change-related objectives of adaptation and mitigation.
- GIZ and other organizations such as the United Nations Development Programme (UNDP), UK Department for International development and The World Bank are providing technical assistance to the interested Indian States in the development of SAPCC.

The State-level plans are not only important for implementation of the NAPCC, but also inclusion of particular regional and local characteristics and specific concerns of vulnerable sectors and communities within each State. Let's now see few examples.

13.4.1 Tamil Nadu

The State of Tamil Nadu in order to further their proactive measures against climate change; and in response to National Action Plan on Climate Change (NAPCC), established the Tamil Nadu State Climate Change Cell

(TNSCCC). It has also established web portal to disseminate information. The cell has visualized for building the capacity of all the stakeholders to respond climate change with various measures. It has a vision to make the state resilient. It has planned to collect necessary data and analyse the same to disseminate scientific information among farmers', fishermen, general public, policy planners', decisions makers', bureaucrats' and others. The important scientific contribution to address climate change was the development of "ClimaRice". "ClimaRice" is indeed an "Intercontinental collaborative project" entitled "Climate Change and persistent Droughts: Impact, vulnerability and adaptation in rice growing sub- divisions in India". Through this project, climate change impacts on Cauvery Basin of Tamil Nadu, which is drought prone have been investigated both in the current and projected climate scenarios. Eventually, a series of adaptation measures have been developed with an aim of sustaining the production in the Cauvery Basin.

The strategies suggested as part of the project to address climate change in the Cauvery basin are:

- Creating awareness among farmers' and other stakeholders' to do minimum tillage to retain soil carbon content;
- Strategies to develop and promote the "use of drought and flood tolerant varieties";
- Promoting the crop rotation practices that augment soil productivity;
- Short duration pulse crops are recommended as relay crop, which would enrich the soil with nitrogen, improve soil fertility, and reduce dependence on the chemical fertilizers and pesticides;
- Encouraging the use of quality inputs like quality seeds, and market intelligence;
- Encouraging the farmers' to grow green manure crops, and use Biofertilizers;
- Introduction of modified cultivation methods such as "System of Rice Intensification"; and
- Promotion of crop residue retention on soil to build up soil biomass.

The broad strategies for climate change adaptation in agriculture and horticulture sector in Tamil Nadu are given in the box 13.1.

Box 13.1: Broad strategies for climate change adaptation in agriculture and horticulture sectors in Tamil Nadu

- Sowing is recommended to be in consonance with the onset of South-west and North- east monsoons;
- Soil erosion must be managed through measures such as minimum tillage, mulching, etc.;
- Soil health and soil nutrient content may be improved through soil- test based Integrated Nutrient Management;
- Promotion of Integrated Pest and Disease Management;
- Water-use efficiency can be improved through methods such as drip,

sprinkler irrigation, etc.;

- System of Rice Intensification (SRI) can double or triple current rice yields;
- Emphasis must be on to develop and introduce new crop varieties that are tolerant to high temperature and water stress;
- “Crop Diversification” is recommended to augment the livelihoods;
- Integrated Farming System is recommended to utilize the farm resources efficiently.
- Risk Mitigation Measures such as compulsory crop insurance are recommended to compensate for the crop losses due to extreme weather events;
- Supply of quality agricultural inputs;
- Capacity building on latest agricultural technologies;
- “Rain water management practices at household and landscape level”;
- “Institute a long-term rice varietal development program to make available new rice varieties and conserving local gene pool through public-private partnership, and actively engaging local farmers, women, and other local agencies in the process”.
- “Help farmers adopt tailored farming practices including soil nutrient management, choice of crop and cultivars, and pests and disease management as per the suitability/potential of different agro climatic zones based on weather based advisories derived from climate analysis of the local area and forecasting which includes probable occurrence of pests and disease”.
- “Support research on weather-based crop insurance schemes to cover risks of increasing intensities and frequencies of extreme weather events. The challenge is to develop a weather-based index to simplify the insurance claims and payment procedures”.

Reference:<http://www.environment.tn.gov.in/doc/TNSAPCC%20PDF/Chapter%205%20Sustainable%20Agriculture%20.pdf>; Government of Tamil Nadu. 2013. State Action Plan on Climate Change. Chennai: Government of Tamil Nadu.

13.4.2 Delhi

Cities in India are expected to grow with approximately additional 500 million populations. In this transition, governments at all levels especially city government need to equip themselves to address the needs of population in the areas of physical infrastructure, social infrastructure, housing, water, public transportation, health, education, sewage system and solid waste management. Cities, especially mega cities in India with high concentration of population are at risk from the impact of climate and related issues.

Delhi, the capital of India, with geographical area of 1483 square km with high air pollution has developed specific agenda to address and combat climate change. Delhi face threat from climate change and related issues on

infrastructure, human lives, human health, personal property, environmental quality and future prosperity. Delhi has introduced a Delhi Climate Change Agenda (2009–2012) by incorporating actions to address the issues raised in the Prime Minister's National Action Plan on Climate Change. Delhi Government's agenda aims to reduce Delhi's carbon footprint by identifying 65 set of action points. These actions will be addressed by taking appropriate measures at the policy level by the various departments of Delhi government. For example, the government of Delhi introduced compressed natural gas (CNG) fuelled buses for public transport by replacing existing diesel fuelled buses. Delhi government with the help of central government implemented Jawaharlal Nehru Urban Renewal Mission (JNURM) in 2005 itself to address issues related to urban development. But these schemes should be integrated with climate adaptation, mitigation and climate change related risk management.

Delhi cabinet has decided to upgrade energy efficiency of existing government buildings through retrofitting which are to be carried out by energy service companies in a performance contracting mode. The objective is to ensure that the government buildings can achieve at least rating of one star from BEE under their office building labelling programme. Delhi government is also considering the installation of energy efficient water and wastewater treatment operations and methane recovery and reuse for a CDM project.

13.4.3 Jharkhand

Jharkhand has area coverage of 79,714 square km. Jharkhand has 24 administrative districts. Ranchi is the state capital as well as an industrial city. It has rich physiographic characteristics. The rivers Swarnrekha, Koyal, Shankh and Damodar flow through the State and are utilized for irrigation in many places in the State. Jharkhand has around 40 per cent of the country's mineral resources such as coal, iron ore, copper, uranium, mica, bauxite, granite, limestone, silver, graphite, magnetite and dolomite. It possess about 40 per cent of the country's mineral wealth. 11 per cent to the State's GSDP comes from mining and quarrying activities. The State mineral reserves include coal, iron ore, bauxite, copper, mica, graphite, manganese, lead, silver, uranium and limestone. The State exports minerals to Bangladesh, Nepal, South Africa and Saudi Arabia. As regards the measures with respect to climate change mitigation, the state has proposed a scheme called Jharkhand Solar Policy (2015).

Jharkhand Solar Policy 2015

The objective of the scheme is to encourage participation of private sector to set up solar power based projects in the State and increase solar power generation to 2500 MW by the year 2020 in a phased manner.

The Jharkhand Renewable Energy Development Agency (JREDA) was incorporated as a society in the year 2001 for promoting use of renewable energy sources in the state. Being a nodal agency, JREDA is working for implementation of fiscal and financial incentives made available by the Ministry of New and Renewable Energy Sources (MNRES), Govt. of India and Indian Renewable Energy Development Agency (IREDA).

13.5 ASSESSMENT OF STATE ACTION PLANS ON CLIMATE CHANGE

This section broadly assesses the existing State Action Plans to address climate change. India has to prepare detailed nation-wide project for 15 agro-climatic zones with varying vulnerabilities. By considering the vastness and varying agro climatic zones, central government made all States to participate actively to address climate change by assessing their own vulnerabilities. It needs to prepare detailed plan document on par with NAPCC. It needs to identify projects and prepare budgeting and monitoring mechanisms. State governments have freedom to involve local self-governments to implement projects related to climate change. This exercise was considered as one of the biggest in the world. The State Action Committees submitted 32 plan proposals to the Union Ministry of Environment, Forest and Climate Change (MoEF & CC). The Common framework document for SAPCCs circulated by MoEF & CC requires States to assess “the physical and economic impact of and vulnerability to climate change on the most vulnerable sectors and vulnerable groups”. This is very important to plan, adopt and implement mitigation strategies. For example, States like Tamil Nadu, Gujarat, Maharashtra, Karnataka, Orissa, West Bengal and Andhra Pradesh have long coast lines. These States are vulnerable to frequent cyclone and flood. Fishermen and cities close to coastal areas get affected by the unexpected monsoons. It is necessary for these states to assess vulnerability of fishermen and others who are directly depended on sea for their livelihoods. The analysis of Centre for Science and Environment (CSE) shows that “the SAPCCs submitted by most states lack detailed vulnerability assessments. Some are so broad and general that they risk overlooking specific local issues, while others like Gujarat, Odisha and Tamil Nadu have assessed their vulnerability based on a few projects. SAPCCs of Mizoram and Uttarakhand do not even mention vulnerability assessment”. According to the report prepared by CSE, the Uttarakhand State has conducted several consultations among civil society groups and academics to recognize the severity of climate vulnerabilities. The result of the consultation was missing in the report. Punjab conducted stakeholder consultation for preparing its SAPCC. Mizoram neither held consultations with civil society nor vulnerable communities. Some have adopted international and national level models and used the same to assess and prepare vulnerabilities. But, these models have their own limitations with respect to projects, region or State specific changes and vulnerabilities. India’s climate system is monsoon driven. States must plan adaptation and mitigation strategies according to climate change projection scenarios. According to CSE analysis, “Mizoram and Odisha lack climate projections in their SAPCCs, while others, including Gujarat, rely on climate models used by the UK. Madhya Pradesh made mid-century and end-century projections based on secondary data collected from various sources. CSE researchers found that the states are depending on secondary sources due to lack of domestic climate models. The outcomes of such projections based on flawed models have a degree of uncertainty as climate change impact is highly local in nature” (Down to Earth, 2018).

The next comes finance. The States have no clarity about financing. It has assumption that the central government or International organization would provide required finance to implement their projects. States like Madhya Pradesh demanded Rupees 4,700 crores, while Tamil Nadu demanded more than Rupees 400,000 crores. Gujarat allocated around 80 per cent to the water sector. Most states do not depict the true picture of their vulnerability in the SAPCC document and the required steps that need to be taken to enhance their adaptive capacity and reduce their vulnerability to climate change extremes. The documents lack mentioning the effective monitoring and evaluation institutions for the project implemented to address climate change. States such as Kerala and Uttarakhand, are in the process of revising their SAPCCs to meet their domestic and international objectives after recognising the inadequacy in the existing document. “The state of Mizoram has initiated a ninth mission on health, in addition to the eight national-level missions for climate change. The state government is of the opinion that health is one of the major climate change impacts and hence has included it as an additional mission. The National Adaptation Fund for Climate Change (NAFCC) is funding a project to augment livelihood of rural communities by building resilience in agriculture. The project is being implemented by the state agriculture department and is currently in its third year” (Down to Earth, 2018).

Check Your Progress: 2

Note: 1) Use the space given below for your answers.

2) Check your answers with those given at the end of the unit.

1. Write short note on the policies of the State of Tamil Nadu to mitigate climate change.

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2. What is State Action Plan on Climate Change?

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13.5 LET US SUM UP

In this unit, we have discussed the process of policy formulation. The process certainly help the learners to understand the steps in policy formulation in India. The process is explained in the context of climate change policy. The State Action Plan on Climate Change (SAPCC) of few states have been discussed in the unit.

13.6 KEYWORDS

International Organization : Organization which presents at international level with scope of dealing issues among nations.

Niti Aayog: It is a premier think tank of Government of India to provide critical knowledge for policy formulation.

13.7 FURTHER READING

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Web Links

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<https://www.ipcc.ch/report/managing-the-risks-of-extreme-events-and-disasters-to-advance-climate-change-adaptation/>

<http://www.ipcc.ch/report/ar5/wg1/> <http://www.ipcc.ch/report/ar5/wg2/>

<http://www.ipcc.ch/report/ar5/wg3/>

<http://www.ipcc.ch/report/ar5/syr/> <https://www.ipcc.ch/sr15/>

<http://www.eea.europa.eu/data-and-maps/indicators/atmospheric-greenhouse-gas-concentrations/atmospheric-greenhouse-gas-concentrations-assessment-3>

13.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Policy is defined as “the product of political influence, determining and setting limits to what the state does”.
2. Policy formulation is a process rather than one time affair. There are various stakeholders involved in public policy formulation process like political parties, civil society organizations, individuals, elected

representatives, local bodies apart from executives and representatives of legislatures. Thus, the process of policy formulation includes drafting, consultation, negotiation, reformulation, finalization, legislation, converting policies into programmes to achieve desired results. The success of any public policy depends upon identifying issues, drafting policy, and making all stakeholders to involve in the process actively and converting the policy into programmes for implementation. There are various agencies involved in policy formulation.

In India, both executives and political wings of the government involve in the process. Apart from that, there are large number of agencies/organisations involve in the process of policy formulation directly/indirectly. Informally, civil society groups, individuals, various associations, non-governmental organizations involve in policy formulation in the way of providing suggestions to improve policies.

Check Your Progress: 2

1. The State of Tamil Nadu in order to further their proactive measures against climate change; and in response to National Action Plan on Climate Change (NAPCC), established the Tamil Nadu State Climate Change Cell (TNSCCC). It has also established web portal to disseminate information. The cell has visualized for building the capacity of all the stakeholders to respond climate change with various measures. It has planned to collect necessary data and analyse the same to disseminate scientific information among farmers', fishermen, general public, policy planners', decisions makers', bureaucrats' and others. The important scientific contribution to address climate change was the development of "ClimaRice". "ClimaRice" is indeed an "Intercontinental collaborative project" entitled "Climate Change and persistent Droughts: Impact, vulnerability and adaptation in rice growing sub-divisions in India". Through this project, climate change impacts on Cauvery Basin of Tamil Nadu, which is drought prone have been investigated both in the current and projected climate scenarios. A series of adaptation measures have been developed with an aim of sustaining the production in the Cauvery Basin.
2. Government of India formulated NAPCC and requested all State governments to formulate State Action plan on Climate change (SAPCC). In 2009, the Government of India notified state governments to come up with their own climate change policies related to mitigation and adaptation. The plans and actions should be aligned with their respective state priorities as well as the NAPCC goals. The State Action Plan on Climate Change (SAPCC) in India is the most notable policy reflecting the nature of decentralized climate change mitigation framework of the country. SAPCCs would address the need of the country as well as provide direction to achieve the climate change mitigation and adaptation.